Supplementary Material: Executable Operating Contracts Linking Anytime Localization to Engineering Task Evidence

Chuansu Peng[a], Haoran Dong[a], Ruoxuan Wu[a], Peipei Zeng[b]

[a]*School of Aeronautical Engineering, Civil Aviation University of China, China*
[b]*Engineering Training Center, Civil Aviation University of China, China*

S1. Prospective protocol and execution provenance

This section concerns the R8 prospective study. The later sections labelled retained describe earlier A19/A20, R4 and secondary R5 studies with their original populations, confidence families and timing. They are not pooled with R8. All complete original manuscripts/source archives remain in the release.

Table S1: Disjoint R8 seed blocks. A seed is one independent geometric episode input, not one policy execution.

Block	First seed	Inputs	Role
Excluded pilot	8,200,000	48	Characterization/software audit
Nominal calibration	8,300,000	4,096	Cards and catalogue selection
Unfitted calibration	8,400,000	4,096	Separate cards, unchanged predictor
Nominal confirmation	8,500,000	2,048	Independent frozen comparisons
Unfitted confirmation	8,600,000	2,048	Independent transfer comparisons
Occlusion stress	8,700,000	384	Unsupported descriptive stress
Degeneracy stress	8,800,000	384	Unsupported descriptive stress
Prior-shift stress	8,900,000	384	Unsupported descriptive stress
Map-shift stress	9,000,000	384	Unsupported descriptive stress

Nominal and unfitted populations are uniform discrete mixtures of their listed geometry, prior deviation, noise and discrepancy settings, with independent perturbation/noise draws per seed. A certificate concerns the entire mixture, not each constituent subgroup. Repeated source frames are permitted draws from the fixed route-template population; independently realized sensor/prior/motion perturbations remain the experimental unit. Geometry and template hashes define conditional simulation artifacts rather than a census of physical aircraft variability.

S1.1. F0, attachment amendment and F1

F0 at 2026-10-11 19:07:18.664676 UTC binds 48 numerical/model/artifact components plus the protocol, characterized clock and preflight audit. No final input existed at F0. Frozen predictors are retained CART/RF exports fitted on the earlier 926/194 source-grouped train/validation split; no new input refits either predictor. A19/A20 and R4/R5 data are separate retained studies.

The source-history guard is an explicitly post-F0, pre-F1 attachment amendment. V1 (19:33:16 UTC) demonstrated numerical noninterference but allowed a freshly reconstructed procedural ledger to reset the same source's latch. Independent statistical review identified this discrepancy. V2 introduced persistent scope state; further provenance/timer edits were completed before a final V3 seal at 19:46:40.196132 UTC. Intermediate manifests, matching snapshots, review evidence and superseded status are retained. None was used to generate confirmation data or select a policy. F0, the deterministic event map and family size were unchanged. V3 independently seals the observer, its verification script, integrated confirmation driver and F1 binder.

F1 at 19:59:17.439444 UTC seals 528 real calibration cards, all 96 request plans, calibration means, runtime bindings and calibration-result/input hashes. A preconfirmation inventory at 19:59:32.211933 UTC verified exactly 8,192 raw calibration files and no confirmation/stress files; the separate F1–V3 binding followed at 19:59:32.903035 UTC. Confirmation generation began afterward. These local hashes and execution logs expose chronology but are not external preregistration. The gate-correction note recorded 26 route files present when that later correction was made, a progress snapshot rather than the preconfirmation or final inventory; the final release contains all 36. The executed gate used an incorrect singular route filename; its original source and inventory are preserved, the future glob is corrected, and the independent generation ledger establishes actual route chronology. Two missing unfitted-confirmation JSONL records were restored by rerunning the unchanged collector on their original seeds; all 15 arrays and full NPZ hashes match preserved originals exactly. Their lost original prefix CPU metadata is replaced by recovery-time CPU measurement, not claimed to be bytewise recovered. No source was omitted or added.

S1.2. Exact binomial family and sensitivity

For each of 22 complete policies, two profiles, four services and three events, one independent source contributes one binary failure. The corrected one-sided tail is $\beta/K = 0.05/528$. For $k < n$, inversion of the binomial lower-tail equation gives $U = \mathrm{Beta}^{-1}(1 - \beta/K; k + 1, n - k)$; for $k = n$, $U = 1$. For $k = 0$, $U = 1 - (\beta/K)^{1/n}$. The card loader recomputes the inversion and checks finite bounds, integer counts, K, β and sealed body identity.

Cross-policy/service/event dependence does not reduce the source count and is allowed by the union-bound proof. Four requested tasks and three allowances query the same pre-frozen event cards and add no new event family. The union event is evaluated directly from the two binary failures. Two marginals cannot be substituted for that union guarantee. No route failure, fault-injection fixture or receipt-conditioned subset is added to these denominators.

Nested calibration prefixes 128/256/512/1024/2048/4096 and confidence 0.90/0.95/0.99 recompute diagnostic bounds for the unchanged 528-member family. They do not change F1 or optimize the final sample size. Raw and summary tables contain count, upper bound, requested allowance, signed margin and certified-policy count for every combination. Negative margin means refusal. A small positive margin is not silently promoted to robust support.

S1.3. Work-clock versus native CPU

The pilot characterizes deterministic CPU-equivalent operation coefficients. Solver initialization, correspondence calls and checkpoint costs fit counted native prefixes; foreground voxel processing uses $n \log_2(1+n)$, forecast work counts tree traversals, and propagation/publication/checking primitives are separately timed. The fitted clock is conditioned on its fixed pilot estimate during calibration. It is a simulated service convention with mean costs and timestamp quantization, not measured embedded latency or worst-case execution time.

The hard sender budget includes source seals, ICP/features, forecasting, rule evaluation and publication witnesses. A common next-stage reservation assumes at most six correspondence queries per new iteration plus checkpoint/model/rule/publication work. Frontend and stream/receiver work are outside that stage cap but inside total work and availability. If the next stage is unaffordable, the last eligible candidate may publish; otherwise the service retains its prior. This wrapper changes even fixed-cap behavior, so all cards are newly calibrated effective policies.

Controlled policy traces execute one common native ICP prefix per input and charge its selected prefix to each counterfactual. Rule decisions, source sealing, producer witnesses, validation and motion replay execute for each service/policy. The integrated wrapper's component ledger adds measured claim bookkeeping; its additional inclusive replay ledger includes offline truth assertions and scoring and is explicitly not pure online CPU. Metadata-only R4/R5 microbenchmarks measure different operations and are not added as an end-to-end improvement.

For complete routes, the entire online loop is measured directly, including array copies, frontend, registration/features, forecasting, precommit/dispatch, producer/receiver checks, replay and scheduling. A

versioned input-provider wrapper hydrates compressed raw arrays before the original online timer, avoiding lazy archive decompression inside service timing. Numerical arrays and the frozen route executor are unchanged. A paired provider audit checks complete numeric/clock/event traces. Startup and archive-hydration CPU are recorded separately in worker manifests. CPU is native process time on the documented host; total simulated route duration is not a throughput or embedded real-time measurement.

S1.4. Inference and descriptive analyses

The 16 primary intervals use standard service and both confirmation profiles, loss and observation failure, for Full–validated Flat and KG1–FlatCART/FlatRF/Fixed2. The Full/Flat request is union at allowance 0.15. A refused source retains its prior and stays in the denominator. With 32,000 paired bootstrap replicates, the pointwise interval uses quantiles 0.025/0.975 and the family-adjusted interval uses $0.05/(2 \times 16)$ and its complement. Both are approximate percentile intervals, unlike exact binomial calibration. Input seeds, not repeated policies or time samples, are resampled.

Stress, subgroups, Pareto fronts, route replicates, influence deletion and tail order statistics are descriptive. The 36 route inputs have four paired realizations per geometry/condition with predeclared discrepancy/noise combinations; they are not assumed identically distributed replicates of one mixture. Raw route points and paired replicate differences remain available rather than receiving episode-based confidence claims. The diagnostic route is fixed to unfitted geometry/contention/replicate zero, not selected for a favorable curve.

S1.5. Mechanism assumptions and failure conditions

Source replacement requires a common immutable prior, a retained solve and no interleaved independent update. Attachment assumes trusted producer records and startup card validation, matched active dependencies, pre-sensing commitment and canonical modeled delivery/replay. The history monitor covers rejected operational version attempts associated with the trusted source scope. A normal deadline expiry, empty catalogue or ordered rollback is not a history fault. Once faulted, the source cannot reset its attachment by reconstructing a plan or ledger. Its independent successor scope can still operate.

Fingerprint equality does not detect physical drift, authenticate compromised code or validate a sampling mixture from metadata alone. Undetected missing packets, future motion substituted for causal replay, physical processor delays, changed model/service semantics and external interleaved measurements require revised assumptions or calibration. Remaining authorized receipts do not define a conditionally calibrated subpopulation. The original candidate descriptors in the F0 trace remain reproducibility records; only V3 final receipts are the guarded public attachment.

S1.6. Dependency-lifetime software fixtures

The excluded preflight changes one active dependency at a time and queries the same 528-card registry. Replacing the RF forecast revokes 168 cards, nominal geometry artifacts 264, the union-event definition 176, standard-service definition 132 and the common clock 528. These are constructed dependency tests, not new population failures or a claim of superiority over a procedural implementation with the same dependency validation. Previously serialized assessment objects remain historical snapshots; eligibility must be checked again at current issuance/use.

S2. Prospective result tables and additional figures

All tables below are derived from completed frozen R8 records. Controlled policy outcomes share independent inputs; integrated task executions and route variants are paired repetitions. Counts and units are kept separate from the later retained historical study. Complete precision, every price variant and all CPU/event fields are available in the CSV tables and raw journals.

S2.1. All primary calibration cards

Each row has n=4,096 for each event and K=528. T/O/U mean trajectory/observation/direct union; margins use allowance 0.15. No card is removed because its margin is negative. All policies are freshly calibrated under the common budget wrapper.

Table S2: Nominal calibration: failure counts, exact corrected upper bounds and union margin.

Service	Policy	k_T	k_O	k_U	U_T	U_O	U_U	Margin
relaxed	KG0.3	228	237	283	0.07022	0.07266	0.08503	+0.06497
relaxed	KG1	244	229	278	0.07455	0.07049	0.08369	+0.06631
relaxed	KG3	281	230	285	0.08450	0.07077	0.08557	+0.06443
relaxed	FlatCART0.3	228	237	283	0.07022	0.07266	0.08503	+0.06497
relaxed	FlatCART	244	229	278	0.07455	0.07049	0.08369	+0.06631
relaxed	FlatCART3	281	230	285	0.08450	0.07077	0.08557	+0.06443
relaxed	FlatRF0.3	217	226	267	0.06724	0.06968	0.08074	+0.06926
relaxed	FlatRF	244	220	269	0.07455	0.06805	0.08128	+0.06872
relaxed	FlatRF3	283	231	286	0.08503	0.07104	0.08584	+0.06416
relaxed	ContractRF0.3	217	226	267	0.06724	0.06968	0.08074	+0.06926
relaxed	ContractRF	244	220	269	0.07455	0.06805	0.08128	+0.06872
relaxed	ContractRF3	283	231	286	0.08503	0.07104	0.08584	+0.06416
relaxed	Fixed2	333	284	338	0.09835	0.08530	0.09968	+0.05032
relaxed	Fixed4	216	257	298	0.06697	0.07805	0.08904	+0.06096
relaxed	Fixed8	216	257	298	0.06697	0.07805	0.08904	+0.06096
relaxed	Fixed16	216	257	298	0.06697	0.07805	0.08904	+0.06096
relaxed	Final	216	257	298	0.06697	0.07805	0.08904	+0.06096
relaxed	Residual	331	278	332	0.09782	0.08369	0.09809	+0.05191
relaxed	Random2or4	275	266	310	0.08289	0.08047	0.09224	+0.05776
relaxed	OldRAVC	279	227	283	0.08396	0.06995	0.08503	+0.06497
relaxed	CART_Time	244	229	278	0.07455	0.07049	0.08369	+0.06631
relaxed	RF_Time	244	220	269	0.07455	0.06805	0.08128	+0.06872
standard	KG0.3	226	211	263	0.06968	0.06560	0.07967	+0.07033
standard	KG1	242	209	263	0.07401	0.06506	0.07967	+0.07033
standard	KG3	281	228	286	0.08450	0.07022	0.08584	+0.06416
standard	FlatCART0.3	226	211	263	0.06968	0.06560	0.07967	+0.07033
standard	FlatCART	242	209	263	0.07401	0.06506	0.07967	+0.07033
standard	FlatCART3	281	228	286	0.08450	0.07022	0.08584	+0.06416
standard	FlatRF0.3	333	285	341	0.09835	0.08557	0.10048	+0.04952
standard	FlatRF	333	285	341	0.09835	0.08557	0.10048	+0.04952
standard	FlatRF3	333	285	341	0.09835	0.08557	0.10048	+0.04952
standard	ContractRF0.3	333	285	341	0.09835	0.08557	0.10048	+0.04952
standard	ContractRF	333	285	341	0.09835	0.08557	0.10048	+0.04952
standard	ContractRF3	333	285	341	0.09835	0.08557	0.10048	+0.04952
standard	Fixed2	333	285	341	0.09835	0.08557	0.10048	+0.04952
standard	Fixed4	214	239	284	0.06642	0.07320	0.08530	+0.06470
standard	Fixed8	214	239	284	0.06642	0.07320	0.08530	+0.06470
standard	Fixed16	214	239	284	0.06642	0.07320	0.08530	+0.06470
standard	Final	214	239	284	0.06642	0.07320	0.08530	+0.06470
standard	Residual	331	278	335	0.09782	0.08369	0.09889	+0.05111
standard	Random2or4	275	258	307	0.08289	0.07832	0.09144	+0.05856
standard	OldRAVC	279	226	285	0.08396	0.06968	0.08557	+0.06443
standard	CART_Time	242	209	263	0.07401	0.06506	0.07967	+0.07033
standard	RF_Time	333	285	341	0.09835	0.08557	0.10048	+0.04952
tight	KG0.3	536	3054	3079	0.15152	0.77053	0.77641	-0.62641
tight	KG1	536	3054	3079	0.15152	0.77053	0.77641	-0.62641

Service	Policy	k_T	k_O	k_U	U_T	U_O	U_U	Margin
tight	KG3	536	3054	3079	0.15152	0.77053	0.77641	-0.62641
tight	FlatCART0.3	536	3054	3079	0.15152	0.77053	0.77641	-0.62641
tight	FlatCART	536	3054	3079	0.15152	0.77053	0.77641	-0.62641
tight	FlatCART3	536	3054	3079	0.15152	0.77053	0.77641	-0.62641
tight	FlatRF0.3	536	3959	3961	0.15152	0.97607	0.97648	-0.82648
tight	FlatRF	536	3959	3961	0.15152	0.97607	0.97648	-0.82648
tight	FlatRF3	536	3959	3961	0.15152	0.97607	0.97648	-0.82648
tight	ContractRF0.3	536	3959	3961	0.15152	0.97607	0.97648	-0.82648
tight	ContractRF	536	3959	3961	0.15152	0.97607	0.97648	-0.82648
tight	ContractRF3	536	3959	3961	0.15152	0.97607	0.97648	-0.82648
tight	Fixed2	536	2983	3008	0.15152	0.75378	0.75968	-0.60968
tight	Fixed4	536	2983	3008	0.15152	0.75378	0.75968	-0.60968
tight	Fixed8	536	2983	3008	0.15152	0.75378	0.75968	-0.60968
tight	Fixed16	536	2983	3008	0.15152	0.75378	0.75968	-0.60968
tight	Final	536	2983	3008	0.15152	0.75378	0.75968	-0.60968
tight	Residual	536	2983	3008	0.15152	0.75378	0.75968	-0.60968
tight	Random2or4	536	2983	3008	0.15152	0.75378	0.75968	-0.60968
tight	OldRAVC	536	3035	3060	0.15152	0.76605	0.77194	-0.62194
tight	CART_Time	536	3054	3079	0.15152	0.77053	0.77641	-0.62641
tight	RF_Time	536	3959	3961	0.15152	0.97607	0.97648	-0.82648
contended	KG0.3	536	2081	2124	0.15152	0.53729	0.54775	-0.39775
contended	KG1	536	2081	2124	0.15152	0.53729	0.54775	-0.39775
contended	KG3	536	2081	2124	0.15152	0.53729	0.54775	-0.39775
contended	FlatCART0.3	352	2162	2208	0.10339	0.55698	0.56814	-0.41814
contended	FlatCART	351	2162	2208	0.10312	0.55698	0.56814	-0.41814
contended	FlatCART3	345	2162	2203	0.10154	0.55698	0.56693	-0.41693
contended	FlatRF0.3	368	2763	2789	0.10762	0.70157	0.70777	-0.55777
contended	FlatRF	368	2763	2789	0.10762	0.70157	0.70777	-0.55777
contended	FlatRF3	368	2763	2789	0.10762	0.70157	0.70777	-0.55777
contended	ContractRF0.3	536	2729	2760	0.15152	0.69347	0.70086	-0.55086
contended	ContractRF	536	2729	2760	0.15152	0.69347	0.70086	-0.55086
contended	ContractRF3	536	2729	2760	0.15152	0.69347	0.70086	-0.55086
contended	Fixed2	371	4096	4096	0.10841	1.00000	1.00000	-0.85000
contended	Fixed4	1083	4096	4096	0.29078	1.00000	1.00000	-0.85000
contended	Fixed8	1083	4096	4096	0.29078	1.00000	1.00000	-0.85000
contended	Fixed16	1083	4096	4096	0.29078	1.00000	1.00000	-0.85000
contended	Final	1083	4096	4096	0.29078	1.00000	1.00000	-0.85000
contended	Residual	340	2053	2092	0.10021	0.53048	0.53997	-0.38997
contended	Random2or4	718	4096	4096	0.19835	1.00000	1.00000	-0.85000
contended	OldRAVC	423	2415	2438	0.12208	0.61817	0.62371	-0.47371
contended	CART_Time	536	2081	2124	0.15152	0.53729	0.54775	-0.39775
contended	RF_Time	536	2729	2760	0.15152	0.69347	0.70086	-0.55086

Table S3: Unfitted calibration: failure counts, exact corrected upper bounds and union margin.

Service	Policy	k_T	k_O	k_U	U_T	U_O	U_U	Margin
relaxed	KG0.3	346	310	360	0.10180	0.09224	0.10550	+0.04450
relaxed	KG1	364	329	377	0.10656	0.09729	0.10999	+0.04001
relaxed	KG3	407	351	408	0.11788	0.10312	0.11814	+0.03186
relaxed	FlatCART0.3	346	310	360	0.10180	0.09224	0.10550	+0.04450
relaxed	FlatCART	364	329	377	0.10656	0.09729	0.10999	+0.04001

Service	Policy	k_T	k_O	k_U	U_T	U_O	U_U	Margin
relaxed	FlatCART3	407	351	408	0.11788	0.10312	0.11814	+0.03186
relaxed	FlatRF0.3	344	320	360	0.10127	0.09490	0.10550	+0.04450
relaxed	FlatRF	365	329	373	0.10682	0.09729	0.10893	+0.04107
relaxed	FlatRF3	415	359	416	0.11998	0.10524	0.12024	+0.02976
relaxed	ContractRF0.3	344	320	360	0.10127	0.09490	0.10550	+0.04450
relaxed	ContractRF	365	329	373	0.10682	0.09729	0.10893	+0.04107
relaxed	ContractRF3	415	359	416	0.11998	0.10524	0.12024	+0.02976
relaxed	Fixed2	441	387	442	0.12679	0.11262	0.12705	+0.02295
relaxed	Fixed4	343	323	368	0.10101	0.09570	0.10762	+0.04238
relaxed	Fixed8	343	323	368	0.10101	0.09570	0.10762	+0.04238
relaxed	Fixed16	343	323	368	0.10101	0.09570	0.10762	+0.04238
relaxed	Final	343	323	368	0.10101	0.09570	0.10762	+0.04238
relaxed	Residual	441	387	442	0.12679	0.11262	0.12705	+0.02295
relaxed	Random2or4	382	354	395	0.11131	0.10392	0.11473	+0.03527
relaxed	OldRAVC	396	341	397	0.11499	0.10048	0.11525	+0.03475
relaxed	CART_Time	364	329	377	0.10656	0.09729	0.10999	+0.04001
relaxed	RF_Time	365	329	373	0.10682	0.09729	0.10893	+0.04107
standard	KG0.3	345	310	357	0.10154	0.09224	0.10471	+0.04529
standard	KG1	363	327	374	0.10630	0.09676	0.10920	+0.04080
standard	KG3	407	349	410	0.11788	0.10259	0.11867	+0.03133
standard	FlatCART0.3	345	310	357	0.10154	0.09224	0.10471	+0.04529
standard	FlatCART	363	327	374	0.10630	0.09676	0.10920	+0.04080
standard	FlatCART3	407	349	410	0.11788	0.10259	0.11867	+0.03133
standard	FlatRF0.3	441	384	444	0.12679	0.11183	0.12758	+0.02242
standard	FlatRF	441	384	444	0.12679	0.11183	0.12758	+0.02242
standard	FlatRF3	441	384	444	0.12679	0.11183	0.12758	+0.02242
standard	ContractRF0.3	441	384	444	0.12679	0.11183	0.12758	+0.02242
standard	ContractRF	441	384	444	0.12679	0.11183	0.12758	+0.02242
standard	ContractRF3	441	384	444	0.12679	0.11183	0.12758	+0.02242
standard	Fixed2	441	384	444	0.12679	0.11183	0.12758	+0.02242
standard	Fixed4	340	321	363	0.10021	0.09517	0.10630	+0.04370
standard	Fixed8	340	321	363	0.10021	0.09517	0.10630	+0.04370
standard	Fixed16	340	321	363	0.10021	0.09517	0.10630	+0.04370
standard	Final	340	321	363	0.10021	0.09517	0.10630	+0.04370
standard	Residual	441	384	444	0.12679	0.11183	0.12758	+0.02242
standard	Random2or4	381	352	395	0.11104	0.10339	0.11473	+0.03527
standard	OldRAVC	396	339	397	0.11499	0.09995	0.11525	+0.03475
standard	CART_Time	363	327	374	0.10630	0.09676	0.10920	+0.04080
standard	RF_Time	441	384	444	0.12679	0.11183	0.12758	+0.02242
tight	KG0.3	564	3875	3881	0.15876	0.95829	0.95959	-0.80959
tight	KG1	564	3875	3881	0.15876	0.95829	0.95959	-0.80959
tight	KG3	564	3875	3881	0.15876	0.95829	0.95959	-0.80959
tight	FlatCART0.3	564	3875	3881	0.15876	0.95829	0.95959	-0.80959
tight	FlatCART	564	3875	3881	0.15876	0.95829	0.95959	-0.80959
tight	FlatCART3	564	3875	3881	0.15876	0.95829	0.95959	-0.80959
tight	FlatRF0.3	564	4091	4092	0.15876	0.99989	0.99994	-0.84994
tight	FlatRF	564	4091	4092	0.15876	0.99989	0.99994	-0.84994
tight	FlatRF3	564	4091	4092	0.15876	0.99989	0.99994	-0.84994
tight	ContractRF0.3	564	4091	4092	0.15876	0.99989	0.99994	-0.84994
tight	ContractRF	564	4091	4092	0.15876	0.99989	0.99994	-0.84994
tight	ContractRF3	564	4091	4092	0.15876	0.99989	0.99994	-0.84994
tight	Fixed2	564	3865	3871	0.15876	0.95614	0.95743	-0.80743
tight	Fixed4	564	3865	3871	0.15876	0.95614	0.95743	-0.80743
tight	Fixed8	564	3865	3871	0.15876	0.95614	0.95743	-0.80743
tight	Fixed16	564	3865	3871	0.15876	0.95614	0.95743	-0.80743

Service	Policy	k_T	k_O	k_U	U_T	U_O	U_U	Margin
tight	Final	564	3865	3871	0.15876	0.95614	0.95743	-0.80743
tight	Residual	564	3865	3871	0.15876	0.95614	0.95743	-0.80743
tight	Random2or4	564	3865	3871	0.15876	0.95614	0.95743	-0.80743
tight	OldRAVC	564	3874	3880	0.15876	0.95808	0.95937	-0.80937
tight	CART_Time	564	3875	3881	0.15876	0.95829	0.95959	-0.80959
tight	RF_Time	564	4091	4092	0.15876	0.99989	0.99994	-0.84994
contended	KG0.3	564	3187	3210	0.15876	0.80176	0.80714	-0.65714
contended	KG1	564	3187	3210	0.15876	0.80176	0.80714	-0.65714
contended	KG3	564	3187	3210	0.15876	0.80176	0.80714	-0.65714
contended	FlatCART0.3	442	3264	3295	0.12705	0.81974	0.82696	-0.67696
contended	FlatCART	441	3264	3295	0.12679	0.81974	0.82696	-0.67696
contended	FlatCART3	443	3264	3291	0.12731	0.81974	0.82603	-0.67603
contended	FlatRF0.3	441	3807	3811	0.12679	0.94349	0.94436	-0.79436
contended	FlatRF	441	3807	3811	0.12679	0.94349	0.94436	-0.79436
contended	FlatRF3	441	3807	3811	0.12679	0.94349	0.94436	-0.79436
contended	ContractRF0.3	564	3796	3805	0.15876	0.94107	0.94305	-0.79305
contended	ContractRF	564	3796	3805	0.15876	0.94107	0.94305	-0.79305
contended	ContractRF3	564	3796	3805	0.15876	0.94107	0.94305	-0.79305
contended	Fixed2	442	4096	4096	0.12705	1.00000	1.00000	-0.85000
contended	Fixed4	1157	4096	4096	0.30934	1.00000	1.00000	-0.85000
contended	Fixed8	1157	4096	4096	0.30934	1.00000	1.00000	-0.85000
contended	Fixed16	1157	4096	4096	0.30934	1.00000	1.00000	-0.85000
contended	Final	1157	4096	4096	0.30934	1.00000	1.00000	-0.85000
contended	Residual	441	3153	3183	0.12679	0.79379	0.80082	-0.65082
contended	Random2or4	780	4096	4096	0.21417	1.00000	1.00000	-0.85000
contended	OldRAVC	516	3403	3414	0.14633	0.85200	0.85454	-0.70454
contended	CART_Time	564	3187	3210	0.15876	0.80176	0.80714	-0.65714
contended	RF_Time	564	3796	3805	0.15876	0.94107	0.94305	-0.79305

S2.2. All task requests at the primary allowance

Table S4: All 32 primary-allowance profile/service/task requests. Full and independently implemented validated Flat return identical choices and witnesses.

Profile	Service	Task	Policy	Eligible	Max U	Margin
nominal	relaxed	trajectory	FlatCART3	22	0.084497	+0.065503
nominal	relaxed	observation	FlatCART3	22	0.070765	+0.079235
nominal	relaxed	both_marginals	FlatCART3	22	0.084497	+0.065503
nominal	relaxed	union	FlatCART3	22	0.085568	+0.064432
nominal	standard	trajectory	FlatCART3	22	0.084497	+0.065503
nominal	standard	observation	FlatCART3	22	0.070223	+0.079777
nominal	standard	both_marginals	FlatCART3	22	0.084497	+0.065503
nominal	standard	union	FlatCART3	22	0.085836	+0.064164
nominal	tight	trajectory	Refused	0	–	–
nominal	tight	observation	Refused	0	–	–
nominal	tight	both_marginals	Refused	0	–	–
nominal	tight	union	Refused	0	–	–
nominal	contended	trajectory	FlatCART3	9	0.101536	+0.048464
nominal	contended	observation	Refused	0	–	–
nominal	contended	both_marginals	Refused	0	–	–
nominal	contended	union	Refused	0	–	–
unfitted	relaxed	trajectory	OldRAVC	22	0.114992	+0.035008

Profile	Service	Task	Policy	Eligible	Max U	Margin
unfitted	relaxed	observation	OldRAVC	22	0.100476	+0.049524
unfitted	relaxed	both_marginals	OldRAVC	22	0.114992	+0.035008
unfitted	relaxed	union	OldRAVC	22	0.115254	+0.034746
unfitted	standard	trajectory	OldRAVC	22	0.114992	+0.035008
unfitted	standard	observation	OldRAVC	22	0.099946	+0.050054
unfitted	standard	both_marginals	OldRAVC	22	0.114992	+0.035008
unfitted	standard	union	OldRAVC	22	0.115254	+0.034746
unfitted	tight	trajectory	Refused	0	–	–
unfitted	tight	observation	Refused	0	–	–
unfitted	tight	both_marginals	Refused	0	–	–
unfitted	tight	union	Refused	0	–	–
unfitted	contended	trajectory	OldRAVC	9	0.146330	+0.003670
unfitted	contended	observation	Refused	0	–	–
unfitted	contended	both_marginals	Refused	0	–	–
unfitted	contended	union	Refused	0	–	–

S2.3. Complete confirmation policy catalogue

Table S5: Nominal confirmation, 2,048 inputs/context. L: 10^{-3} m^{2}s; W: total modeled ms/input; success/failure percentages.

Service	Policy	L	W (ms)	Success	T fail	O fail	U fail
relaxed	KG0.3	11.002	11.389	94.14	5.37	5.86	6.88
relaxed	KG1	11.408	10.619	94.24	5.76	5.76	6.93
relaxed	KG3	12.031	9.646	94.19	6.74	5.81	7.08
relaxed	FlatCART0.3	11.002	11.389	94.14	5.37	5.86	6.88
relaxed	FlatCART	11.408	10.619	94.24	5.76	5.76	6.93
relaxed	FlatCART3	12.031	9.646	94.19	6.74	5.81	7.08
relaxed	FlatRF0.3	10.706	12.446	93.95	5.42	6.05	7.13
relaxed	FlatRF	11.125	10.990	94.34	5.86	5.66	6.64
relaxed	FlatRF3	12.152	10.031	94.19	6.88	5.81	7.23
relaxed	ContractRF0.3	10.706	12.446	93.95	5.42	6.05	7.13
relaxed	ContractRF	11.125	10.990	94.34	5.86	5.66	6.64
relaxed	ContractRF3	12.152	10.031	94.19	6.88	5.81	7.23
relaxed	Fixed2	13.830	9.221	92.92	8.06	7.08	8.59
relaxed	Fixed4	10.911	12.072	93.21	5.52	6.79	7.81
relaxed	Fixed8	10.911	12.072	93.21	5.52	6.79	7.81
relaxed	Fixed16	10.911	12.072	93.21	5.52	6.79	7.81
relaxed	Final	10.911	12.072	93.21	5.52	6.79	7.81
relaxed	Residual	13.590	9.403	93.12	8.11	6.88	8.45
relaxed	Random2or4	12.254	10.507	92.58	6.98	7.42	8.50
relaxed	OldRAVC	11.950	9.600	94.58	6.40	5.42	6.74
relaxed	CART_Time	11.408	10.619	94.24	5.76	5.76	6.93
relaxed	RF_Time	11.125	10.990	94.34	5.86	5.66	6.64
standard	KG0.3	11.269	11.495	94.38	5.32	5.62	6.79
standard	KG1	11.664	10.705	94.48	5.71	5.52	6.84
standard	KG3	12.301	9.721	94.24	6.74	5.76	7.13
standard	FlatCART0.3	11.269	11.495	94.38	5.32	5.62	6.79
standard	FlatCART	11.664	10.705	94.48	5.71	5.52	6.84
standard	FlatCART3	12.301	9.721	94.24	6.74	5.76	7.13
standard	FlatRF0.3	13.844	9.973	92.82	8.06	7.18	8.64
standard	FlatRF	13.844	9.973	92.82	8.06	7.18	8.64
standard	FlatRF3	13.844	9.973	92.82	8.06	7.18	8.64

Service	Policy	L	W (ms)	Success	T fail	O fail	U fail
standard	ContractRF0.3	13.844	9.973	92.82	8.06	7.18	8.64
standard	ContractRF	13.844	9.973	92.82	8.06	7.18	8.64
standard	ContractRF3	13.844	9.973	92.82	8.06	7.18	8.64
standard	Fixed2	13.833	9.221	92.82	8.06	7.18	8.64
standard	Fixed4	11.295	12.109	93.51	5.47	6.49	7.52
standard	Fixed8	11.295	12.109	93.51	5.47	6.49	7.52
standard	Fixed16	11.295	12.109	93.51	5.47	6.49	7.52
standard	Final	11.295	12.109	93.51	5.47	6.49	7.52
standard	Residual	13.817	9.476	93.02	8.11	6.98	8.50
standard	Random2or4	12.379	10.520	92.87	6.93	7.13	8.25
standard	OldRAVC	12.199	9.666	94.58	6.40	5.42	6.79
standard	CART_Time	11.664	10.705	94.48	5.71	5.52	6.84
standard	RF_Time	13.844	9.973	92.82	8.06	7.18	8.64
tight	KG0.3	17.401	7.522	26.56	10.69	73.44	73.63
tight	KG1	17.401	7.522	26.56	10.69	73.44	73.63
tight	KG3	17.401	7.522	26.56	10.69	73.44	73.63
tight	FlatCART0.3	17.401	7.522	26.56	10.69	73.44	73.63
tight	FlatCART	17.401	7.522	26.56	10.69	73.44	73.63
tight	FlatCART3	17.401	7.522	26.56	10.69	73.44	73.63
tight	FlatRF0.3	17.474	7.784	3.56	10.69	96.44	96.44
tight	FlatRF	17.474	7.784	3.56	10.69	96.44	96.44
tight	FlatRF3	17.474	7.784	3.56	10.69	96.44	96.44
tight	ContractRF0.3	17.474	7.784	3.56	10.69	96.44	96.44
tight	ContractRF	17.474	7.784	3.56	10.69	96.44	96.44
tight	ContractRF3	17.474	7.784	3.56	10.69	96.44	96.44
tight	Fixed2	17.391	7.511	28.27	10.69	71.73	71.92
tight	Fixed4	17.391	7.511	28.27	10.69	71.73	71.92
tight	Fixed8	17.391	7.511	28.27	10.69	71.73	71.92
tight	Fixed16	17.391	7.511	28.27	10.69	71.73	71.92
tight	Final	17.391	7.511	28.27	10.69	71.73	71.92
tight	Residual	17.391	7.511	28.27	10.69	71.73	71.92
tight	Random2or4	17.391	7.511	28.27	10.69	71.73	71.92
tight	OldRAVC	17.397	7.519	27.15	10.69	72.85	73.05
tight	CART_Time	17.401	7.522	26.56	10.69	73.44	73.63
tight	RF_Time	17.474	7.784	3.56	10.69	96.44	96.44
contended	KG0.3	17.620	7.583	50.78	10.69	49.22	49.95
contended	KG1	17.620	7.583	50.78	10.69	49.22	49.95
contended	KG3	17.620	7.583	50.78	10.69	49.22	49.95
contended	FlatCART0.3	12.170	12.053	49.12	8.40	50.88	52.49
contended	FlatCART	12.527	11.183	49.12	8.30	50.88	52.39
contended	FlatCART3	13.098	10.098	49.12	7.86	50.88	52.05
contended	FlatRF0.3	14.577	10.351	33.15	8.84	66.85	67.82
contended	FlatRF	14.577	10.351	33.15	8.84	66.85	67.82
contended	FlatRF3	14.577	10.351	33.15	8.84	66.85	67.82
contended	ContractRF0.3	17.679	7.839	33.84	10.69	66.16	66.60
contended	ContractRF	17.679	7.839	33.84	10.69	66.16	66.60
contended	ContractRF3	17.679	7.839	33.84	10.69	66.16	66.60
contended	Fixed2	14.802	9.432	0.00	9.23	100.00	100.00
contended	Fixed4	12.823	12.377	0.00	24.56	100.00	100.00
contended	Fixed8	12.823	12.377	0.00	24.56	100.00	100.00
contended	Fixed16	12.823	12.377	0.00	24.56	100.00	100.00
contended	Final	12.823	12.377	0.00	24.56	100.00	100.00
contended	Residual	14.481	9.838	51.22	8.40	48.78	50.00
contended	Random2or4	13.608	10.756	0.00	15.38	100.00	100.00
contended	OldRAVC	13.106	10.007	43.46	9.77	56.54	57.03

Service	Policy	L	W (ms)	Success	T fail	O fail	U fail
contended	CART_Time	17.620	7.583	50.78	10.69	49.22	49.95
contended	RF_Time	17.679	7.839	33.84	10.69	66.16	66.60

Table S6: Unfitted confirmation, 2,048 inputs/context. L: 10^{-3} m^{2}s; W: total modeled ms/input; success/failure percentages.

Service	Policy	L	W (ms)	Success	T fail	O fail	U fail
relaxed	KG0.3	16.859	10.998	91.80	8.50	8.20	8.89
relaxed	KG1	17.095	10.598	91.99	8.59	8.01	8.79
relaxed	KG3	18.129	9.667	91.46	9.23	8.54	9.23
relaxed	FlatCART0.3	16.859	10.998	91.80	8.50	8.20	8.89
relaxed	FlatCART	17.095	10.598	91.99	8.59	8.01	8.79
relaxed	FlatCART3	18.129	9.667	91.46	9.23	8.54	9.23
relaxed	FlatRF0.3	16.671	11.917	91.75	8.30	8.25	8.94
relaxed	FlatRF	17.086	11.054	91.65	8.69	8.35	8.94
relaxed	FlatRF3	18.503	10.063	91.11	9.62	8.89	9.62
relaxed	ContractRF0.3	16.671	11.917	91.75	8.30	8.25	8.94
relaxed	ContractRF	17.086	11.054	91.65	8.69	8.35	8.94
relaxed	ContractRF3	18.503	10.063	91.11	9.62	8.89	9.62
relaxed	Fixed2	19.937	9.219	90.87	10.01	9.13	10.06
relaxed	Fixed4	17.030	11.688	91.26	8.45	8.74	9.47
relaxed	Fixed8	17.030	11.688	91.26	8.45	8.74	9.47
relaxed	Fixed16	17.030	11.688	91.26	8.45	8.74	9.47
relaxed	Final	17.030	11.688	91.26	8.45	8.74	9.47
relaxed	Residual	19.609	9.388	91.02	9.91	8.98	9.91
relaxed	Random2or4	18.535	10.348	90.87	9.57	9.13	10.06
relaxed	OldRAVC	17.717	9.718	91.65	9.03	8.35	9.03
relaxed	CART_Time	17.095	10.598	91.99	8.59	8.01	8.79
relaxed	RF_Time	17.086	11.054	91.65	8.69	8.35	8.94
standard	KG0.3	17.194	11.113	91.85	8.50	8.15	8.89
standard	KG1	17.423	10.702	92.09	8.59	7.91	8.79
standard	KG3	18.431	9.750	91.46	9.23	8.54	9.23
standard	FlatCART0.3	17.194	11.113	91.85	8.50	8.15	8.89
standard	FlatCART	17.423	10.702	92.09	8.59	7.91	8.79
standard	FlatCART3	18.431	9.750	91.46	9.23	8.54	9.23
standard	FlatRF0.3	19.936	9.953	90.92	10.01	9.08	10.06
standard	FlatRF	19.936	9.953	90.92	10.01	9.08	10.06
standard	FlatRF3	19.936	9.953	90.92	10.01	9.08	10.06
standard	ContractRF0.3	19.936	9.953	90.92	10.01	9.08	10.06
standard	ContractRF	19.936	9.953	90.92	10.01	9.08	10.06
standard	ContractRF3	19.936	9.953	90.92	10.01	9.08	10.06
standard	Fixed2	19.938	9.220	90.92	10.01	9.08	10.06
standard	Fixed4	17.350	11.736	91.46	8.40	8.54	9.28
standard	Fixed8	17.350	11.736	91.46	8.40	8.54	9.28
standard	Fixed16	17.350	11.736	91.46	8.40	8.54	9.28
standard	Final	17.350	11.736	91.46	8.40	8.54	9.28
standard	Residual	19.857	9.469	91.06	9.91	8.94	9.91
standard	Random2or4	18.675	10.375	90.97	9.52	9.03	9.96
standard	OldRAVC	17.964	9.781	91.60	9.03	8.40	9.03
standard	CART_Time	17.423	10.702	92.09	8.59	7.91	8.79
standard	RF_Time	19.936	9.953	90.92	10.01	9.08	10.06
tight	KG0.3	22.172	7.552	5.13	14.36	94.87	95.12

Service	Policy	L	W (ms)	Success	T fail	O fail	U fail
tight	KG1	22.172	7.552	5.13	14.36	94.87	95.12
tight	KG3	22.172	7.552	5.13	14.36	94.87	95.12
tight	FlatCART0.3	22.172	7.552	5.13	14.36	94.87	95.12
tight	FlatCART	22.172	7.552	5.13	14.36	94.87	95.12
tight	FlatCART3	22.172	7.552	5.13	14.36	94.87	95.12
tight	FlatRF0.3	22.195	7.795	0.05	14.36	99.95	99.95
tight	FlatRF	22.195	7.795	0.05	14.36	99.95	99.95
tight	FlatRF3	22.195	7.795	0.05	14.36	99.95	99.95
tight	ContractRF0.3	22.195	7.795	0.05	14.36	99.95	99.95
tight	ContractRF	22.195	7.795	0.05	14.36	99.95	99.95
tight	ContractRF3	22.195	7.795	0.05	14.36	99.95	99.95
tight	Fixed2	22.168	7.542	5.47	14.36	94.53	94.78
tight	Fixed4	22.168	7.542	5.47	14.36	94.53	94.78
tight	Fixed8	22.168	7.542	5.47	14.36	94.53	94.78
tight	Fixed16	22.168	7.542	5.47	14.36	94.53	94.78
tight	Final	22.168	7.542	5.47	14.36	94.53	94.78
tight	Residual	22.168	7.542	5.47	14.36	94.53	94.78
tight	Random2or4	22.168	7.542	5.47	14.36	94.53	94.78
tight	OldRAVC	22.171	7.549	5.22	14.36	94.78	95.02
tight	CART_Time	22.172	7.552	5.13	14.36	94.87	95.12
tight	RF_Time	22.195	7.795	0.05	14.36	99.95	99.95
contended	KG0.3	22.353	7.616	22.75	14.36	77.25	77.98
contended	KG1	22.353	7.616	22.75	14.36	77.25	77.98
contended	KG3	22.353	7.616	22.75	14.36	77.25	77.98
contended	FlatCART0.3	17.967	11.653	20.85	10.55	79.15	79.64
contended	FlatCART	18.173	11.199	20.85	10.35	79.15	79.69
contended	FlatCART3	19.078	10.148	20.85	10.11	79.15	79.64
contended	FlatRF0.3	20.523	10.344	7.28	10.01	92.72	92.92
contended	FlatRF	20.523	10.344	7.28	10.01	92.72	92.92
contended	FlatRF3	20.523	10.344	7.28	10.01	92.72	92.92
contended	ContractRF0.3	22.405	7.869	7.57	14.36	92.43	92.77
contended	ContractRF	22.405	7.869	7.57	14.36	92.43	92.77
contended	ContractRF3	22.405	7.869	7.57	14.36	92.43	92.77
contended	Fixed2	20.763	9.453	0.00	10.01	100.00	100.00
contended	Fixed4	18.559	12.001	0.00	29.25	100.00	100.00
contended	Fixed8	18.559	12.001	0.00	29.25	100.00	100.00
contended	Fixed16	18.559	12.001	0.00	29.25	100.00	100.00
contended	Final	18.559	12.001	0.00	29.25	100.00	100.00
contended	Residual	20.417	9.853	23.19	10.01	76.81	77.44
contended	Random2or4	19.659	10.623	0.00	19.29	100.00	100.00
contended	OldRAVC	18.611	10.139	17.09	11.96	82.91	83.06
contended	CART_Time	22.353	7.616	22.75	14.36	77.25	77.98
contended	RF_Time	22.405	7.869	7.57	14.36	92.43	92.77

S2.4. Module attribution, including binding stress

Table S7: Matched numerical modules: changed stopping count / input count and paired changes in loss (10^{-3} m^{2}s), work (ms) and observation failure (pp). Profile changes are zero throughout this study; those rows are retained.

Stratum	Service	Module	Changed	n	ΔL	ΔW	ΔO
degeneracy	contended	Time/CART	384	384	+6.990	-3.371	-3.12

Stratum	Service	Module	Changed	n	ΔL	ΔW	ΔO
degeneracy	contended	Profile/CART	0	384	+0.000	+0.000	+0.00
degeneracy	contended	Time/RF	384	384	+3.303	-2.519	-3.65
degeneracy	contended	Profile/RF	0	384	+0.000	+0.000	+0.00
degeneracy	relaxed	Time/CART	0	384	+0.000	+0.000	+0.00
degeneracy	relaxed	Profile/CART	0	384	+0.000	+0.000	+0.00
degeneracy	relaxed	Time/RF	0	384	+0.000	+0.000	+0.00
degeneracy	relaxed	Profile/RF	0	384	+0.000	+0.000	+0.00
degeneracy	standard	Time/CART	0	384	+0.000	+0.000	+0.00
degeneracy	standard	Profile/CART	0	384	+0.000	+0.000	+0.00
degeneracy	standard	Time/RF	0	384	+0.000	+0.000	+0.00
degeneracy	standard	Profile/RF	0	384	+0.000	+0.000	+0.00
degeneracy	tight	Time/CART	0	384	+0.000	+0.000	+0.00
degeneracy	tight	Profile/CART	0	384	+0.000	+0.000	+0.00
degeneracy	tight	Time/RF	0	384	+0.000	+0.000	+0.00
degeneracy	tight	Profile/RF	0	384	+0.000	+0.000	+0.00
geometry	contended	Time/CART	2048	2048	+4.179	-3.583	-1.90
geometry	contended	Profile/CART	0	2048	+0.000	+0.000	+0.00
geometry	contended	Time/RF	2048	2048	+1.882	-2.476	-0.29
geometry	contended	Profile/RF	0	2048	+0.000	+0.000	+0.00
geometry	relaxed	Time/CART	0	2048	+0.000	+0.000	+0.00
geometry	relaxed	Profile/CART	0	2048	+0.000	+0.000	+0.00
geometry	relaxed	Time/RF	0	2048	+0.000	+0.000	+0.00
geometry	relaxed	Profile/RF	0	2048	+0.000	+0.000	+0.00
geometry	standard	Time/CART	0	2048	+0.000	+0.000	+0.00
geometry	standard	Profile/CART	0	2048	+0.000	+0.000	+0.00
geometry	standard	Time/RF	0	2048	+0.000	+0.000	+0.00
geometry	standard	Profile/RF	0	2048	+0.000	+0.000	+0.00
geometry	tight	Time/CART	0	2048	+0.000	+0.000	+0.00
geometry	tight	Profile/CART	0	2048	+0.000	+0.000	+0.00
geometry	tight	Time/RF	0	2048	+0.000	+0.000	+0.00
geometry	tight	Profile/RF	0	2048	+0.000	+0.000	+0.00
map_shift	contended	Time/CART	384	384	-4.215	-4.645	-4.43
map_shift	contended	Profile/CART	0	384	+0.000	+0.000	+0.00
map_shift	contended	Time/RF	384	384	-4.715	-2.470	-1.30
map_shift	contended	Profile/RF	0	384	+0.000	+0.000	+0.00
map_shift	relaxed	Time/CART	0	384	+0.000	+0.000	+0.00
map_shift	relaxed	Profile/CART	0	384	+0.000	+0.000	+0.00
map_shift	relaxed	Time/RF	0	384	+0.000	+0.000	+0.00
map_shift	relaxed	Profile/RF	0	384	+0.000	+0.000	+0.00
map_shift	standard	Time/CART	0	384	+0.000	+0.000	+0.00
map_shift	standard	Profile/CART	0	384	+0.000	+0.000	+0.00
map_shift	standard	Time/RF	0	384	+0.000	+0.000	+0.00
map_shift	standard	Profile/RF	0	384	+0.000	+0.000	+0.00
map_shift	tight	Time/CART	0	384	+0.000	+0.000	+0.00
map_shift	tight	Profile/CART	0	384	+0.000	+0.000	+0.00
map_shift	tight	Time/RF	0	384	+0.000	+0.000	+0.00
map_shift	tight	Profile/RF	0	384	+0.000	+0.000	+0.00
nominal	contended	Time/CART	2048	2048	+5.094	-3.600	-1.66
nominal	contended	Profile/CART	0	2048	+0.000	+0.000	+0.00
nominal	contended	Time/RF	2048	2048	+3.102	-2.511	-0.68
nominal	contended	Profile/RF	0	2048	+0.000	+0.000	+0.00
nominal	relaxed	Time/CART	0	2048	+0.000	+0.000	+0.00
nominal	relaxed	Profile/CART	0	2048	+0.000	+0.000	+0.00
nominal	relaxed	Time/RF	0	2048	+0.000	+0.000	+0.00
nominal	relaxed	Profile/RF	0	2048	+0.000	+0.000	+0.00

Stratum	Service	Module	Changed	n	ΔL	ΔW	ΔO
nominal	standard	Time/CART	0	2048	+0.000	+0.000	+0.00
nominal	standard	Profile/CART	0	2048	+0.000	+0.000	+0.00
nominal	standard	Time/RF	0	2048	+0.000	+0.000	+0.00
nominal	standard	Profile/RF	0	2048	+0.000	+0.000	+0.00
nominal	tight	Time/CART	0	2048	+0.000	+0.000	+0.00
nominal	tight	Profile/CART	0	2048	+0.000	+0.000	+0.00
nominal	tight	Time/RF	0	2048	+0.000	+0.000	+0.00
nominal	tight	Profile/RF	0	2048	+0.000	+0.000	+0.00
occlusion	contended	Time/CART	328	384	+5.777	-3.142	-1.30
occlusion	contended	Profile/CART	0	384	+0.000	+0.000	+0.00
occlusion	contended	Time/RF	328	384	+4.468	-2.159	-0.52
occlusion	contended	Profile/RF	0	384	+0.000	+0.000	+0.00
occlusion	relaxed	Time/CART	0	384	+0.000	+0.000	+0.00
occlusion	relaxed	Profile/CART	0	384	+0.000	+0.000	+0.00
occlusion	relaxed	Time/RF	0	384	+0.000	+0.000	+0.00
occlusion	relaxed	Profile/RF	0	384	+0.000	+0.000	+0.00
occlusion	standard	Time/CART	0	384	+0.000	+0.000	+0.00
occlusion	standard	Profile/CART	0	384	+0.000	+0.000	+0.00
occlusion	standard	Time/RF	0	384	+0.000	+0.000	+0.00
occlusion	standard	Profile/RF	0	384	+0.000	+0.000	+0.00
occlusion	tight	Time/CART	0	384	+0.000	+0.000	+0.00
occlusion	tight	Profile/CART	0	384	+0.000	+0.000	+0.00
occlusion	tight	Time/RF	0	384	+0.000	+0.000	+0.00
occlusion	tight	Profile/RF	0	384	+0.000	+0.000	+0.00
prior_shift	contended	Time/CART	384	384	+44.844	-5.143	+0.00
prior_shift	contended	Profile/CART	0	384	+0.000	+0.000	+0.00
prior_shift	contended	Time/RF	384	384	+36.695	-2.488	+0.00
prior_shift	contended	Profile/RF	0	384	+0.000	+0.000	+0.00
prior_shift	relaxed	Time/CART	0	384	+0.000	+0.000	+0.00
prior_shift	relaxed	Profile/CART	0	384	+0.000	+0.000	+0.00
prior_shift	relaxed	Time/RF	0	384	+0.000	+0.000	+0.00
prior_shift	relaxed	Profile/RF	0	384	+0.000	+0.000	+0.00
prior_shift	standard	Time/CART	0	384	+0.000	+0.000	+0.00
prior_shift	standard	Profile/CART	0	384	+0.000	+0.000	+0.00
prior_shift	standard	Time/RF	0	384	+0.000	+0.000	+0.00
prior_shift	standard	Profile/RF	0	384	+0.000	+0.000	+0.00
prior_shift	tight	Time/CART	0	384	+0.000	+0.000	+0.00
prior_shift	tight	Profile/CART	0	384	+0.000	+0.000	+0.00
prior_shift	tight	Time/RF	0	384	+0.000	+0.000	+0.00
prior_shift	tight	Profile/RF	0	384	+0.000	+0.000	+0.00

S2.5. Integrated CPU and evidence accounting

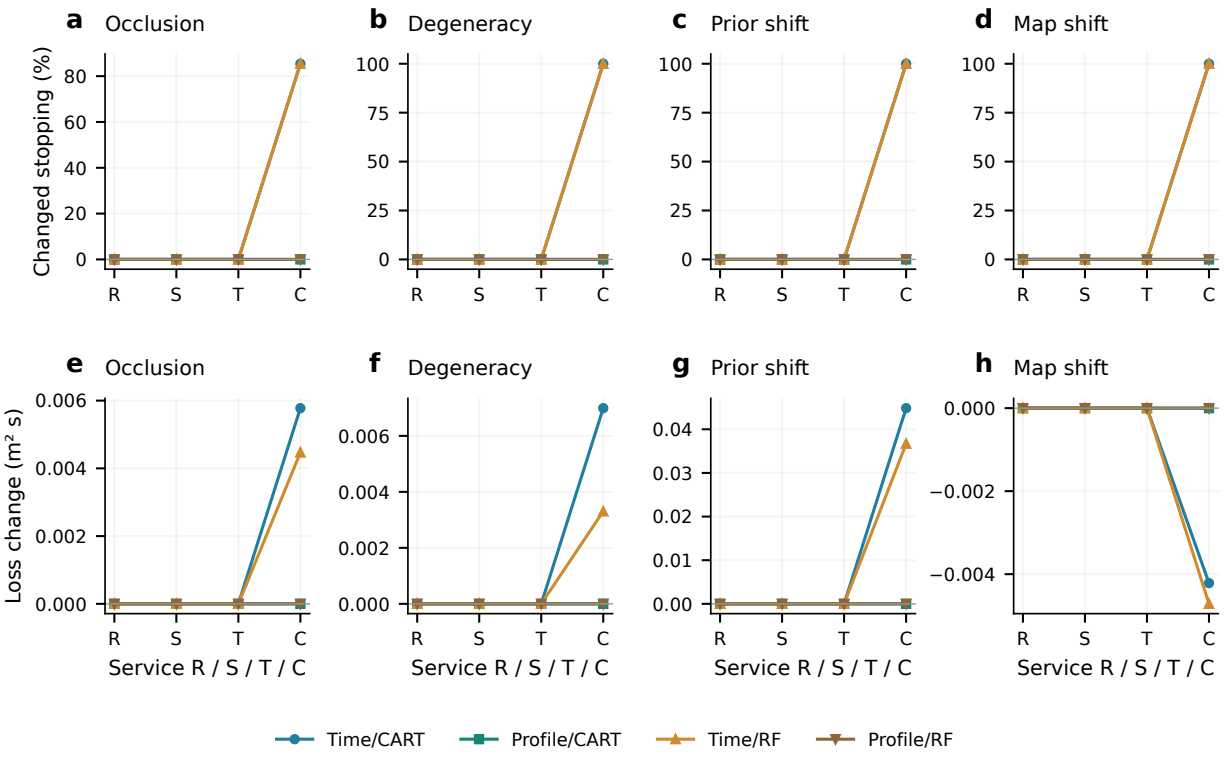


Figure S1: Binding-constraint stress interventions on all 384 inputs/stratum. R/S/T/C denote the four services. Time screening changes many endpoints under contention and can increase or decrease loss; profile addition remains numerically redundant. These unsupported profiles receive no primary risk card.

Table S8: Actual Full confirmation. Coverage requires at least one valid issued receipt during execution for each requested event. Dispatch/extra claim columns are microseconds; component and inclusive program ledgers are milliseconds. Inclusive replay contains offline assertion/scoring and is not pure online CPU. Validated Flat has identical numeric/receipt outcomes and its independent CPU rows remain in integrated_summary.csv.

Profile	Service	Task	Coverage	Dispatch μs	Extra μs	Component	Inclusive
nominal	contended	both_marginals	0.000	53.0	39.4	3.512	3.551
nominal	contended	observation	0.000	44.7	32.5	3.496	3.565
nominal	contended	trajectory	100.000	146.2	93.1	10.683	10.776
nominal	contended	union	0.000	39.6	22.1	3.481	3.529
nominal	relaxed	both_marginals	100.000	196.2	102.8	10.349	10.395
nominal	relaxed	observation	100.000	166.4	102.9	10.219	10.290
nominal	relaxed	trajectory	100.000	233.9	117.8	10.419	10.459
nominal	relaxed	union	100.000	150.5	100.1	10.202	10.284
nominal	standard	both_marginals	100.000	192.1	104.5	10.366	10.413
nominal	standard	observation	100.000	156.6	99.7	10.289	10.369
nominal	standard	trajectory	100.000	186.2	99.6	10.312	10.364
nominal	standard	union	100.000	145.8	100.1	10.289	10.378
nominal	tight	both_marginals	0.000	29.3	22.0	3.470	3.535
nominal	tight	observation	0.000	44.9	24.5	3.489	3.538
nominal	tight	trajectory	0.000	72.0	32.4	3.524	3.587
nominal	tight	union	0.000	40.7	30.1	3.490	3.533
unfitted	contended	both_marginals	0.000	56.7	24.7	3.578	3.617
unfitted	contended	observation	0.000	51.2	32.5	3.581	3.649
unfitted	contended	trajectory	99.951	155.4	97.0	10.700	10.787
unfitted	contended	union	0.000	43.3	23.7	3.564	3.612
unfitted	relaxed	both_marginals	99.951	209.4	108.1	10.319	10.356
unfitted	relaxed	observation	99.951	171.2	107.2	10.238	10.310
unfitted	relaxed	trajectory	99.951	224.7	122.6	10.387	10.466
unfitted	relaxed	union	99.951	155.4	105.5	10.225	10.305
unfitted	standard	both_marginals	99.951	203.8	107.7	10.365	10.404
unfitted	standard	observation	99.951	161.3	105.3	10.382	10.460
unfitted	standard	trajectory	99.951	179.6	105.2	10.320	10.375
unfitted	standard	union	99.951	156.4	104.0	10.304	10.383
unfitted	tight	both_marginals	0.000	32.1	24.2	3.553	3.621
unfitted	tight	observation	0.000	48.0	25.9	3.571	3.625
unfitted	tight	trajectory	0.000	70.0	32.3	3.599	3.665
unfitted	tight	union	0.000	46.6	112.6	3.656	3.703

S2.6. All complete-route methods and strata

Table S9: All twelve live route variants: descriptive means of four matched realizations per geometry/condition. Error units cm; CPU s/600-s route; success %; no route certificate. Four raw replicate differences per comparison remain in route_paired_descriptive.csv.

Geometry	Condition	Method	RMSE	p99	Success	CPU	Switch
B767	ordinary	Fixed2	3.806	9.790	97.07	2.450	0
B767	ordinary	Fixed4	3.719	9.189	94.57	3.176	0
B767	ordinary	Final	3.719	9.189	94.57	3.228	0
B767	ordinary	FlatCART	3.706	8.875	96.49	2.711	0

Geometry	Condition	Method	RMSE	p99	Success	CPU	Switch
B767	ordinary	FlatRF	3.805	9.759	97.07	2.596	0
B767	ordinary	ContractRF	3.805	9.759	97.07	2.728	0
B767	ordinary	OldRAVC	3.722	9.019	97.07	2.500	0
B767	ordinary	KG1	3.706	8.875	96.49	2.809	0
B767	ordinary	CART_Time	3.706	8.875	96.49	2.767	0
B767	ordinary	KG1 held	3.712	8.972	96.32	2.562	0
B767	ordinary	Full	3.715	9.036	97.24	2.514	0
B767	ordinary	Validated Flat	3.715	9.036	97.24	2.504	0
B767	contention	Fixed2	3.555	8.093	46.99	2.322	0
B767	contention	Fixed4	3.605	8.219	45.48	2.607	0
B767	contention	Final	3.605	8.219	45.48	2.566	0
B767	contention	FlatCART	3.581	8.233	54.10	2.505	0
B767	contention	FlatRF	3.549	8.098	44.57	2.576	0
B767	contention	ContractRF	3.523	8.123	44.57	2.345	0
B767	contention	OldRAVC	3.550	8.095	54.85	2.299	0
B767	contention	KG1	3.543	8.059	54.43	2.246	0
B767	contention	CART_Time	3.543	8.059	54.43	2.038	0
B767	contention	KG1 held	3.545	8.065	54.43	2.325	0
B767	contention	Full	3.531	8.090	55.02	2.019	48
B767	contention	Validated Flat	3.531	8.090	55.02	2.158	48
B767	stress	Fixed2	8.185	15.492	42.22	2.154	0
B767	stress	Fixed4	8.215	15.547	39.97	2.557	0
B767	stress	Final	8.215	15.547	39.97	2.668	0
B767	stress	FlatCART	8.215	15.581	53.85	2.583	0
B767	stress	FlatRF	8.187	15.514	47.41	2.450	0
B767	stress	ContractRF	8.186	15.537	47.74	2.136	0
B767	stress	OldRAVC	8.212	15.527	53.18	2.440	0
B767	stress	KG1	8.200	15.384	55.85	2.207	0
B767	stress	CART_Time	8.200	15.384	55.85	2.213	0
B767	stress	KG1 held	8.205	15.384	55.60	2.199	0
B767	stress	Full	8.158	15.541	56.19	2.300	48
B767	stress	Validated Flat	8.158	15.541	56.19	2.175	48
C172	ordinary	Fixed2	4.787	10.404	94.15	2.297	0
C172	ordinary	Fixed4	4.836	11.084	91.64	3.189	0
C172	ordinary	Final	4.836	11.084	91.64	2.941	0
C172	ordinary	FlatCART	4.832	10.478	93.06	2.378	0
C172	ordinary	FlatRF	4.787	10.404	94.15	2.651	0
C172	ordinary	ContractRF	4.787	10.404	94.15	2.675	0
C172	ordinary	OldRAVC	4.837	10.760	93.65	2.267	0
C172	ordinary	KG1	4.832	10.478	93.06	2.589	0
C172	ordinary	CART_Time	4.832	10.478	93.06	2.607	0
C172	ordinary	KG1 held	4.832	10.475	93.06	2.562	0
C172	ordinary	Full	4.805	10.436	93.81	2.518	0
C172	ordinary	Validated Flat	4.805	10.436	93.81	2.491	0
C172	contention	Fixed2	5.237	10.604	53.01	2.121	0
C172	contention	Fixed4	5.388	11.451	51.76	2.659	0
C172	contention	Final	5.388	11.451	51.76	2.734	0
C172	contention	FlatCART	5.321	10.828	81.02	2.374	0
C172	contention	FlatRF	5.236	10.604	53.68	2.288	0
C172	contention	ContractRF	5.256	10.616	53.93	2.090	0
C172	contention	OldRAVC	5.193	10.582	81.77	2.173	0
C172	contention	KG1	5.341	10.842	80.94	2.194	0
C172	contention	CART_Time	5.341	10.842	80.94	2.162	0
C172	contention	KG1 held	5.342	10.843	80.85	2.096	0
C172	contention	Full	5.234	10.569	81.61	2.028	48

Geometry	Condition	Method	RMSE	p99	Success	CPU	Switch
C172	contention	Validated Flat	5.234	10.569	81.61	2.230	48
C172	stress	Fixed2	14.488	28.613	10.20	2.168	0
C172	stress	Fixed4	14.333	27.971	10.03	2.545	0
C172	stress	Final	14.333	27.971	10.03	2.754	0
C172	stress	FlatCART	14.494	28.595	11.62	2.432	0
C172	stress	FlatRF	14.490	28.606	6.19	2.285	0
C172	stress	ContractRF	14.398	28.363	6.27	2.342	0
C172	stress	OldRAVC	14.403	28.828	11.45	2.480	0
C172	stress	KG1	14.439	28.348	11.45	2.198	0
C172	stress	CART_Time	14.439	28.348	11.45	2.054	0
C172	stress	KG1 held	14.440	28.348	11.45	2.120	0
C172	stress	Full	14.414	28.056	11.20	2.221	48
C172	stress	Validated Flat	14.414	28.056	11.20	2.189	48
Unfitted	ordinary	Fixed2	2.997	6.103	99.41	2.413	0
Unfitted	ordinary	Fixed4	3.022	6.256	98.91	2.688	0
Unfitted	ordinary	Final	3.022	6.256	98.91	2.674	0
Unfitted	ordinary	FlatCART	3.018	6.230	99.25	2.635	0
Unfitted	ordinary	FlatRF	2.997	6.103	99.41	2.521	0
Unfitted	ordinary	ContractRF	2.997	6.103	99.41	2.673	0
Unfitted	ordinary	OldRAVC	3.006	6.137	99.25	2.290	0
Unfitted	ordinary	KG1	3.018	6.230	99.25	2.618	0
Unfitted	ordinary	CART_Time	3.018	6.230	99.25	2.507	0
Unfitted	ordinary	KG1 held	3.027	6.255	99.16	2.577	0
Unfitted	ordinary	Full	3.006	6.137	99.25	2.395	0
Unfitted	ordinary	Validated Flat	3.006	6.137	99.25	2.235	0
Unfitted	contention	Fixed2	3.064	6.134	39.21	2.108	0
Unfitted	contention	Fixed4	3.071	6.206	39.21	2.353	0
Unfitted	contention	Final	3.071	6.206	39.21	2.694	0
Unfitted	contention	FlatCART	3.054	6.153	43.73	2.382	0
Unfitted	contention	FlatRF	3.057	6.111	40.22	2.322	0
Unfitted	contention	ContractRF	2.993	5.868	40.38	2.319	0
Unfitted	contention	OldRAVC	3.048	6.118	43.81	2.383	0
Unfitted	contention	KG1	2.993	5.903	43.65	2.078	0
Unfitted	contention	CART_Time	2.993	5.903	43.65	2.027	0
Unfitted	contention	KG1 held	2.996	5.909	43.65	1.934	0
Unfitted	contention	Full	2.994	5.869	43.65	1.975	48
Unfitted	contention	Validated Flat	2.994	5.869	43.65	2.090	48
Unfitted	stress	Fixed2	9.665	27.479	38.21	2.017	0
Unfitted	stress	Fixed4	9.283	24.449	37.04	2.519	0
Unfitted	stress	Final	9.283	24.449	37.04	2.531	0
Unfitted	stress	FlatCART	9.672	27.692	45.07	2.500	0
Unfitted	stress	FlatRF	9.661	27.564	42.14	2.451	0
Unfitted	stress	ContractRF	9.633	27.118	42.56	1.907	0
Unfitted	stress	OldRAVC	9.269	24.467	44.82	2.464	0
Unfitted	stress	KG1	9.649	27.115	46.49	2.147	0
Unfitted	stress	CART_Time	9.649	27.115	46.49	2.207	0
Unfitted	stress	KG1 held	9.654	27.115	46.49	2.042	0
Unfitted	stress	Full	9.663	27.116	45.74	2.211	48
Unfitted	stress	Validated Flat	9.663	27.116	45.74	2.401	48

S2.7. Tail and influence evidence

Table S10: KG1 influence diagnostics. Gain is baseline minus KG1 (10^{-3} m^2s); top-two share is descriptive. I/W/T counts improved/worsened/tied inputs and retains all n=2,048. No top contributor is excluded from primary inference.

Profile	Service	Baseline	Gain	Top 2 (%)	Rest gain	I/W/T
geometry	contended	Fixed2	-1.589	-22.63	-1.951	851/1194/3
geometry	contended	FlatCART	-4.179	-7.37	-4.492	666/1382/0
geometry	contended	FlatRF	-1.830	-19.66	-2.192	694/1351/3
geometry	relaxed	Fixed2	2.842	8.94	2.591	1456/515/77
geometry	relaxed	FlatCART	0.000	–	0.000	0/0/2048
geometry	relaxed	FlatRF	-0.009	-1137.37	-0.112	277/292/1479
geometry	standard	Fixed2	2.514	10.02	2.265	458/366/1224
geometry	standard	FlatCART	0.000	–	0.000	0/0/2048
geometry	standard	FlatRF	2.512	10.03	2.262	455/365/1228
geometry	tight	Fixed2	-0.004	-0.00	-0.004	0/7/2041
geometry	tight	FlatCART	0.000	–	0.000	0/0/2048
geometry	tight	FlatRF	0.023	19.51	0.019	85/28/1935
nominal	contended	Fixed2	-2.818	-15.79	-3.266	860/1182/6
nominal	contended	FlatCART	-5.094	-2.11	-5.206	750/1298/0
nominal	contended	FlatRF	-3.043	-14.62	-3.492	779/1263/6
nominal	relaxed	Fixed2	2.422	25.72	1.801	1202/637/209
nominal	relaxed	FlatCART	0.000	–	0.000	0/0/2048
nominal	relaxed	FlatRF	-0.283	-33.03	-0.377	271/292/1485
nominal	standard	Fixed2	2.169	27.99	1.563	442/374/1232
nominal	standard	FlatCART	0.000	–	0.000	0/0/2048
nominal	standard	FlatRF	2.180	27.85	1.575	497/373/1178
nominal	tight	Fixed2	-0.011	-2.75	-0.011	4/31/2013
nominal	tight	FlatCART	0.000	–	0.000	0/0/2048
nominal	tight	FlatRF	0.073	5.40	0.069	372/153/1523

Table S11: Standard-service tails. Loss columns use 10^{-3} m^2s; error columns cm. Largest-5% means are descriptive order statistics over every source.

Profile	Policy	L p95	L p99	L top 5%	Error p99	Worst error
nominal	Fixed2	47.252	105.101	95.488	16.862	52.980
nominal	Fixed4	37.359	76.244	65.348	14.654	25.083
nominal	Final	37.359	76.244	65.348	14.654	25.083
nominal	FlatCART	37.550	77.893	67.486	14.654	25.083
nominal	FlatRF	47.252	105.459	95.513	16.862	52.980
nominal	ContractRF	47.252	105.459	95.513	16.862	52.980
nominal	OldRAVC	39.557	87.184	74.421	15.294	25.083
nominal	KG1	37.550	77.893	67.486	14.654	25.083
unfitted	Fixed2	82.222	286.196	215.495	27.364	43.740
unfitted	Fixed4	71.743	239.320	183.938	25.776	44.647
unfitted	Final	71.743	239.320	183.938	25.776	44.647
unfitted	FlatCART	69.775	242.270	187.717	25.776	44.647
unfitted	FlatRF	82.222	286.196	215.467	27.364	43.740
unfitted	ContractRF	82.222	286.196	215.467	27.364	43.740
unfitted	OldRAVC	72.850	246.840	190.171	26.276	44.647
unfitted	KG1	69.775	242.270	187.717	25.776	44.647

S2.8. Audit evidence and retained earlier result displays

The preflight reports 12 cold-solve equivalence cases, 384 live-prefix/service-budget comparisons, 48 full/procedural equivalence cases and 1,632 constructed fault checks, plus 12 invalid card-schema and five dependency-invalidation fixtures. The final V3 audit covers 16,896 paired pilot contexts and 5,632 first-calibration-chunk policy/service projections, 14 named fault/rollback contexts and entry-point restoration. These software counts are not geometric calibration sample sizes. The real-card audit adds 16 invalid and two valid constructed cases: incorrect attachments 16/4/2/0/0 for no checks/matching/current/history/validated procedural. The normal cases pass all layers. Its deterministic scenario selection supports conformance attribution, not an unbiased field-fault prevalence estimate.

The archive-provider audit reexecutes two complete inputs in both modes. All trace/capture arrays and non-CPU jobs/work/events agree exactly. It adds four executions and zero independent route observations; native CPU differences illustrate archive-boundary contamination rather than a method improvement. Historical metadata checker/dispatcher microbenchmarks retain their original measurement scope and are not combined with current online timing. All review defects, corrections, recovery records and original histories remain in the release.

The following valid R4/R5 experimental displays remain visible with their original populations and timing, separate from new confirmation evidence. Supplied simulation screenshots and all four original physical photos appear in the main paper, with originals retained byte-for-byte.

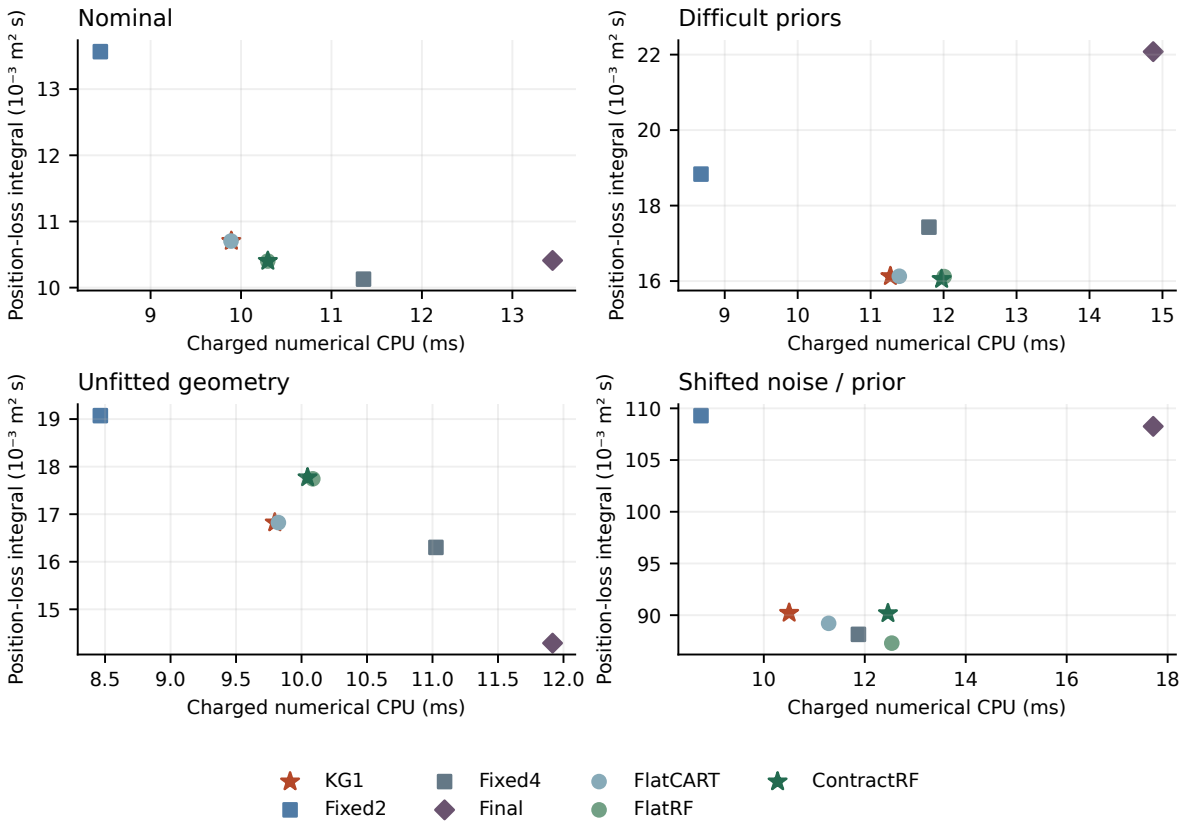


Figure S2: Retained R4 work–loss frontier; Fixed2 and matched Flat controls remain visible.

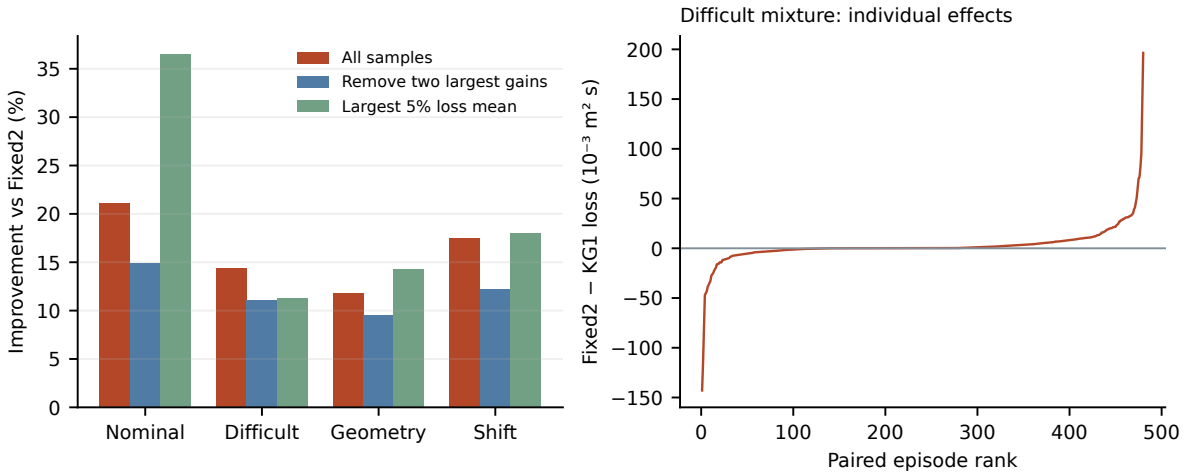


Figure S3: Retained R4 influence diagnostic; its contributor concentration is not substituted for R8.

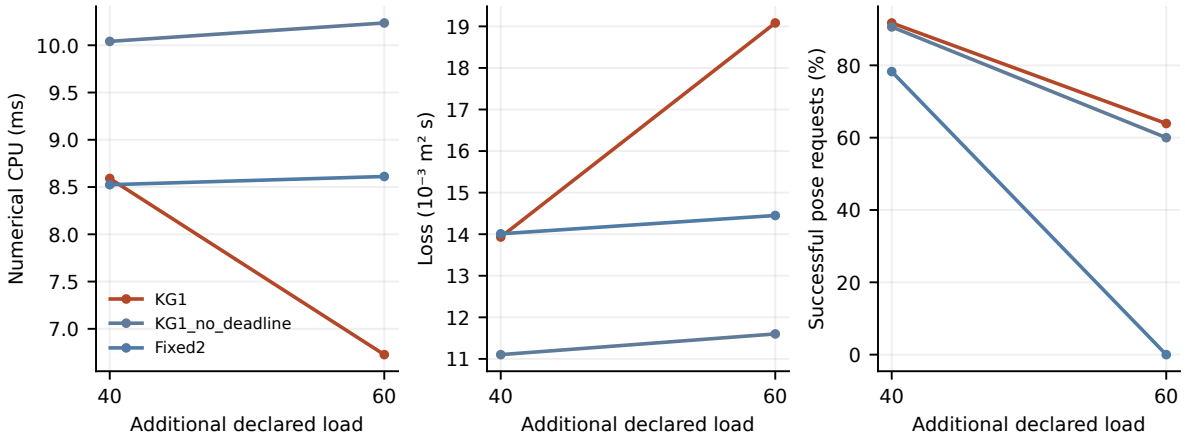


Figure S4: Retained R4 unsupported workload results; no old card is transferred to the new service.

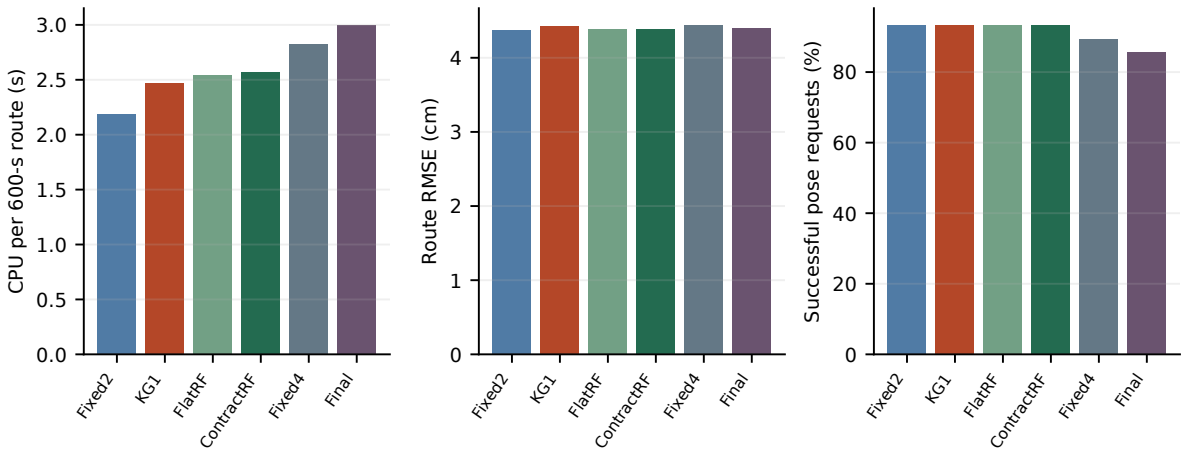


Figure S5: Retained R4 nine-input/81-execution complete routes; these are separate from the 36 new inputs.

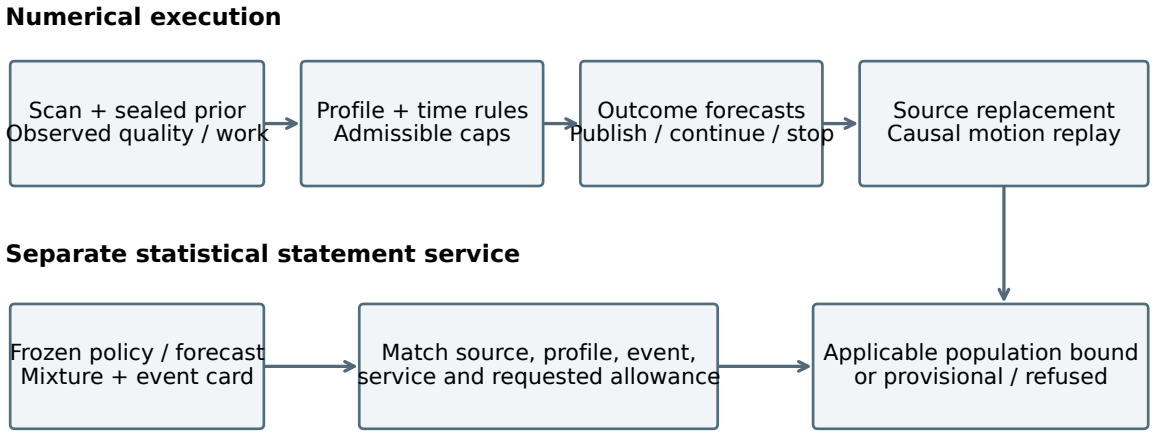


Figure S6: Retained R4 operating-constraint/card comparison with its original finite family.

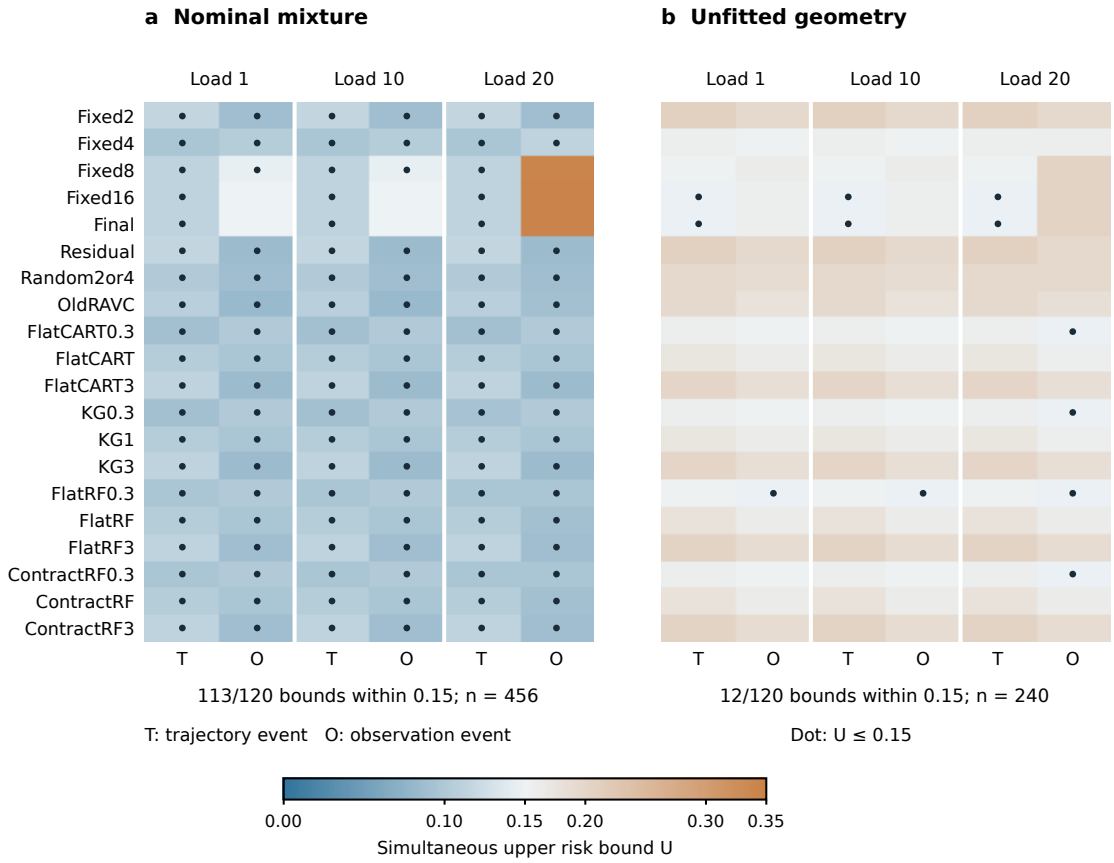


Figure S7: All 240 retained R4 population cards; separate from the 528 new cards.

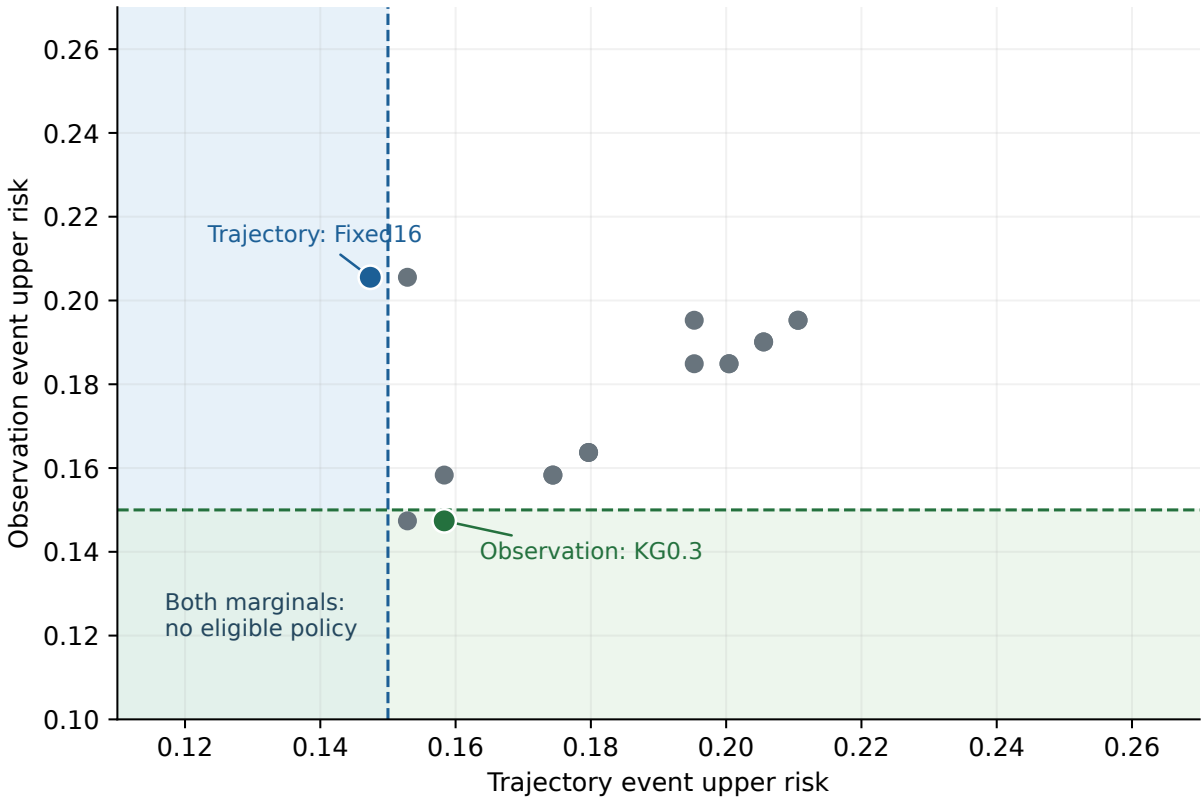


Figure S8: Retained R5 post-primary event-selection analysis; explicitly secondary rather than new confirmation.

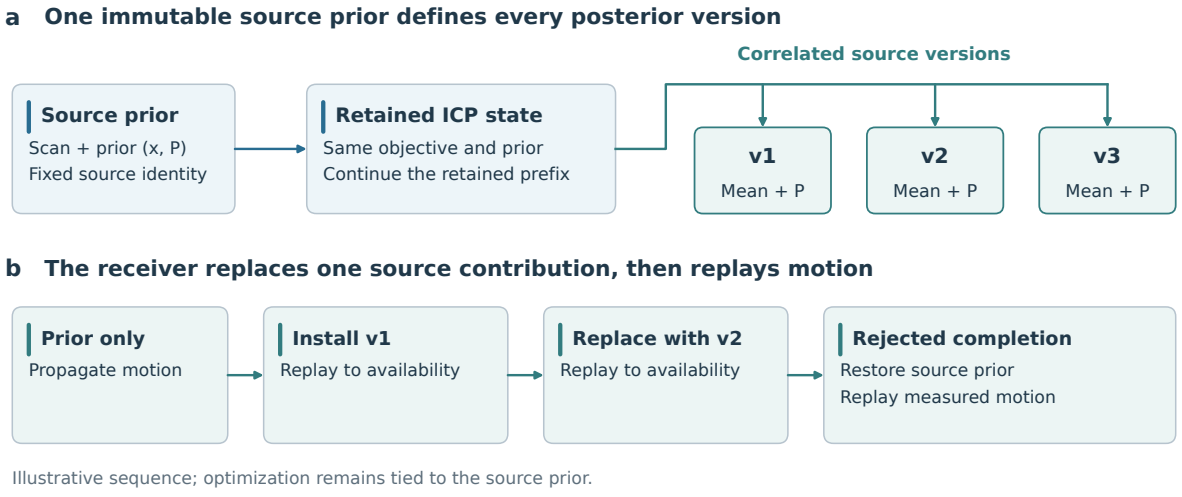


Figure S9: Retained explanatory source-version replacement and replay interface.

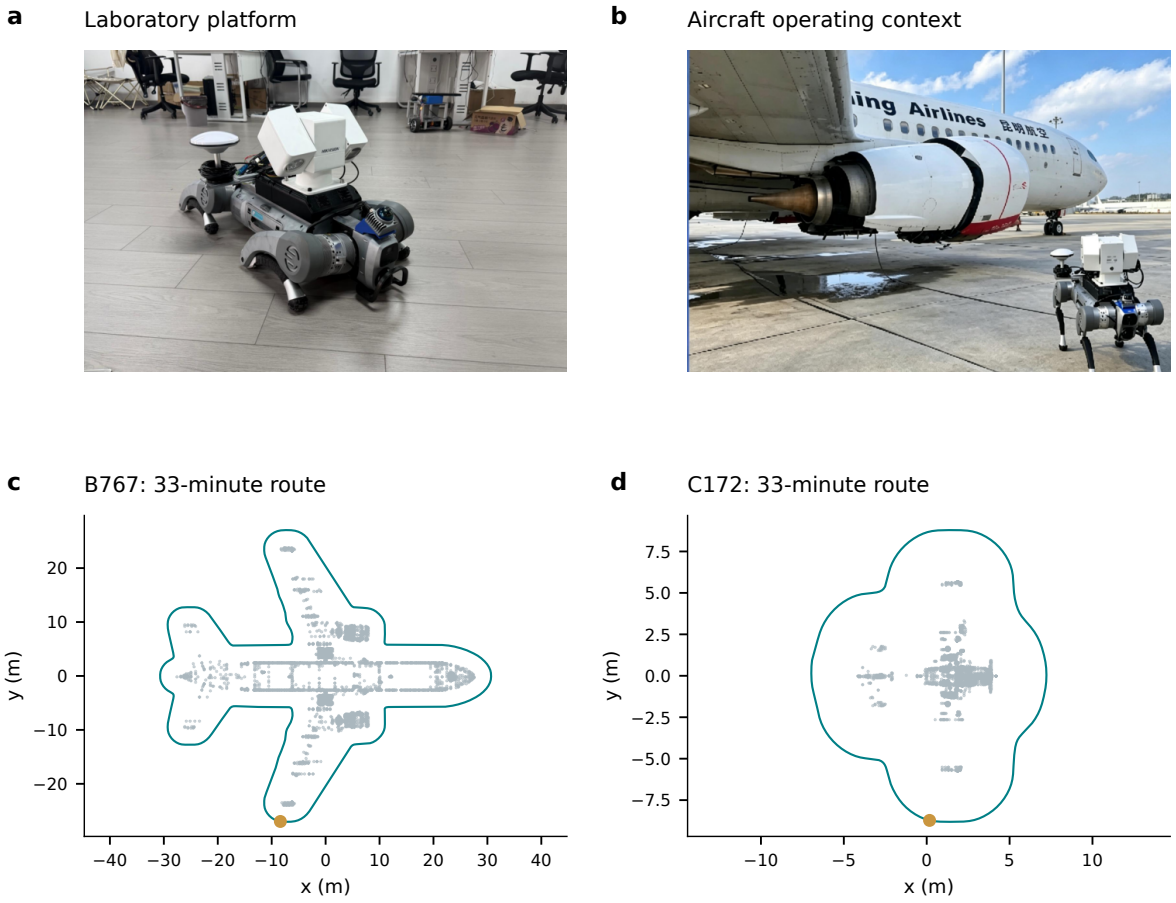


Figure S10: Retained original platform photographs and prescribed aircraft route geometry.

Preserved earlier study: distinct inputs, policies, families and timing

S3. Retained R4 protocol and family records

Final seeds are 5,100,000–5,100,479 (nominal calibration/benchmark), 5,200,000–5,200,639 (nominal confirmation), 5,300,000–5,300,479 (difficult confirmation), 5,400,000–5,400,359 (shift confirmation), 5,500,000–5,500,239 (geometry calibration) and 5,600,000–5,600,479 (geometry confirmation). Benchmark seeds 5,100,000–5,100,023 are excluded from calibration. Ranges are disjoint from fitting, earlier studies and reconstruction pilots.

Tables report the entire 20-policy family. KG is contract-CART; FlatCART/FlatRF omit guards, while ContractRF retains them. Suffix 0.3 or 3 gives price; an omitted suffix means price 1. Residual stops at an eligible second iteration; Random2or4 uses the frozen development continuation frequency. OldRAVC is the earlier frozen policy. Bounds use family size 240 and total confidence error 0.05; allowance 0.15 requires matching event, mixture and service.

Table S12: Nominal calibration at load 1. Event-specific counts and simultaneous upper risks.

Policy	n	k_{traj}	U_{traj}	k_{obs}	U_{obs}	CPU (ms)	L ($\mathrm{m}^2\mathrm{s}$)
ContractRF	456	26	0.10524	23	0.09671	10.073	0.01026
ContractRF0.3	456	23	0.09671	25	0.10241	11.702	0.00967
ContractRF3	456	29	0.11363	20	0.08802	9.195	0.01138
Final	456	29	0.11363	43	0.15149	13.234	0.01055
Fixed16	456	29	0.11363	43	0.15149	13.234	0.01055
Fixed2	456	30	0.11640	20	0.08802	8.456	0.01154
Fixed4	456	23	0.09671	26	0.10524	11.290	0.00978
Fixed8	456	29	0.11363	41	0.14619	13.162	0.01054
FlatCART	456	26	0.10524	23	0.09671	9.745	0.01048
FlatCART0.3	456	21	0.09093	25	0.10241	10.700	0.00994
FlatCART3	456	29	0.11363	19	0.08508	8.839	0.01086
FlatRF	456	26	0.10524	23	0.09671	10.070	0.01026
FlatRF0.3	456	23	0.09671	25	0.10241	11.693	0.00967
FlatRF3	456	29	0.11363	20	0.08802	9.190	0.01138
KG0.3	456	21	0.09093	25	0.10241	10.730	0.00994
KG1	456	26	0.10524	23	0.09671	9.762	0.01048
KG3	456	29	0.11363	19	0.08508	8.848	0.01086
OldRAVC	456	27	0.10805	18	0.08212	8.788	0.01107
Random2or4	456	25	0.10241	20	0.08802	9.796	0.01046
Residual	456	30	0.11640	19	0.08508	8.530	0.01153

Table S13: Nominal calibration at load 10. Event-specific counts and simultaneous upper risks.

Policy	n	k_{traj}	U_{traj}	k_{obs}	U_{obs}	CPU (ms)	L ($\mathrm{m}^2\mathrm{s}$)
ContractRF	456	26	0.10524	23	0.09671	10.169	0.01044
ContractRF0.3	456	23	0.09671	25	0.10241	11.830	0.00986
ContractRF3	456	29	0.11363	20	0.08802	9.270	0.01153
Final	456	29	0.11363	43	0.15149	13.301	0.01105
Fixed16	456	29	0.11363	43	0.15149	13.302	0.01105
Fixed2	456	30	0.11640	20	0.08802	8.516	0.01192
Fixed4	456	23	0.09671	26	0.10524	11.359	0.01028
Fixed8	456	29	0.11363	41	0.14619	13.229	0.01103
FlatCART	456	26	0.10524	23	0.09671	9.837	0.01065
FlatCART0.3	456	21	0.09093	25	0.10241	10.813	0.01013
FlatCART3	456	29	0.11363	19	0.08508	8.911	0.01102
FlatRF	456	26	0.10524	23	0.09671	10.163	0.01044
FlatRF0.3	456	23	0.09671	25	0.10241	11.822	0.00986
FlatRF3	456	29	0.11363	20	0.08802	9.263	0.01153
KG0.3	456	21	0.09093	25	0.10241	10.822	0.01013
KG1	456	26	0.10524	23	0.09671	9.847	0.01065
KG3	456	29	0.11363	19	0.08508	8.918	0.01102
OldRAVC	456	27	0.10805	18	0.08212	8.854	0.01128
Random2or4	456	25	0.10241	20	0.08802	9.858	0.01088
Residual	456	30	0.11640	19	0.08508	8.596	0.01167

Table S14: Nominal calibration at load 20. Event-specific counts and simultaneous upper risks.

Policy	n	k_{traj}	U_{traj}	k_{obs}	U_{obs}	CPU (ms)	L ($\mathrm{m}^2\mathrm{s}$)
ContractRF	456	26	0.10524	21	0.09093	10.231	0.01079
ContractRF0.3	456	23	0.09671	23	0.09671	11.848	0.01023
ContractRF3	456	29	0.11363	20	0.08802	9.329	0.01186
Final	456	29	0.11363	121	0.34343	13.372	0.01162
Fixed16	456	29	0.11363	121	0.34343	13.371	0.01162
Fixed2	456	30	0.11640	20	0.08802	8.527	0.01201
Fixed4	456	23	0.09671	29	0.11363	11.420	0.01074
Fixed8	456	29	0.11363	120	0.34109	13.299	0.01161
FlatCART	456	26	0.10524	23	0.09671	9.932	0.01100
FlatCART0.3	456	21	0.09093	25	0.10241	10.932	0.01048
FlatCART3	456	29	0.11363	19	0.08508	8.985	0.01136
FlatRF	456	26	0.10524	21	0.09093	10.250	0.01078
FlatRF0.3	456	23	0.09671	23	0.09671	11.946	0.01022
FlatRF3	456	29	0.11363	20	0.08802	9.336	0.01185
KG0.3	456	22	0.09383	25	0.10241	10.867	0.01055
KG1	456	26	0.10524	23	0.09671	9.930	0.01100
KG3	456	29	0.11363	19	0.08508	8.980	0.01136
OldRAVC	456	27	0.10805	21	0.09093	8.918	0.01162
Random2or4	456	25	0.10241	20	0.08802	9.895	0.01115
Residual	456	30	0.11640	19	0.08508	8.670	0.01199

Table S15: Procedural geometry calibration at load 1. Event-specific counts and simultaneous upper risks.

Policy	n	k_{traj}	U_{traj}	k_{obs}	U_{obs}	CPU (ms)	L ($\mathrm{m}^2\mathrm{s}$)
ContractRF	240	23	0.17966	20	0.16372	9.728	0.01871
ContractRF0.3	240	19	0.15832	18	0.15288	10.734	0.01778
ContractRF3	240	28	0.20554	25	0.19011	8.912	0.02099
Final	240	17	0.14739	19	0.15832	11.572	0.01597
Fixed16	240	17	0.14739	19	0.15832	11.572	0.01597
Fixed2	240	29	0.21063	26	0.19528	8.223	0.02142
Fixed4	240	19	0.15832	18	0.15288	10.727	0.01779
Fixed8	240	18	0.15288	20	0.16372	11.484	0.01629
FlatCART	240	22	0.17438	20	0.16372	9.568	0.01862
FlatCART0.3	240	19	0.15832	18	0.15288	9.975	0.01792
FlatCART3	240	27	0.20042	24	0.18490	8.642	0.02015
FlatRF	240	23	0.17966	20	0.16372	9.749	0.01871
FlatRF0.3	240	18	0.15288	17	0.14739	10.882	0.01694
FlatRF3	240	28	0.20554	25	0.19011	8.927	0.02099
KG0.3	240	19	0.15832	18	0.15288	9.909	0.01801
KG1	240	22	0.17438	20	0.16372	9.557	0.01862
KG3	240	27	0.20042	24	0.18490	8.625	0.02015
OldRAVC	240	26	0.19528	23	0.17966	8.559	0.01949
Random2or4	240	26	0.19528	25	0.19011	9.412	0.02070
Residual	240	29	0.21063	26	0.19528	8.292	0.02142

Table S16: Procedural geometry calibration at load 10. Event-specific counts and simultaneous upper risks.

Policy	n	k_{traj}	U_{traj}	k_{obs}	U_{obs}	CPU (ms)	L ($\mathrm{m}^2\mathrm{s}$)
ContractRF	240	23	0.17966	20	0.16372	9.822	0.01876
ContractRF0.3	240	19	0.15832	18	0.15288	10.848	0.01786
ContractRF3	240	28	0.20554	25	0.19011	8.984	0.02099
Final	240	17	0.14739	19	0.15832	11.641	0.01622
Fixed16	240	17	0.14739	19	0.15832	11.641	0.01622
Fixed2	240	29	0.21063	26	0.19528	8.278	0.02147
Fixed4	240	19	0.15832	18	0.15288	10.796	0.01798
Fixed8	240	18	0.15288	20	0.16372	11.552	0.01651
FlatCART	240	22	0.17438	20	0.16372	9.661	0.01867
FlatCART0.3	240	19	0.15832	18	0.15288	10.077	0.01800
FlatCART3	240	27	0.20042	24	0.18490	8.713	0.02017
FlatRF	240	23	0.17966	20	0.16372	9.840	0.01876
FlatRF0.3	240	18	0.15288	17	0.14739	10.999	0.01703
FlatRF3	240	28	0.20554	25	0.19011	9.001	0.02099
KG0.3	240	19	0.15832	18	0.15288	9.988	0.01808
KG1	240	22	0.17438	20	0.16372	9.640	0.01868
KG3	240	27	0.20042	24	0.18490	8.694	0.02017
OldRAVC	240	26	0.19528	23	0.17966	8.620	0.01948
Random2or4	240	26	0.19528	25	0.19011	9.472	0.02079
Residual	240	29	0.21063	26	0.19528	8.355	0.02143

Table S17: Procedural geometry calibration at load 20. Event-specific counts and simultaneous upper risks.

Policy	n	k_{traj}	U_{traj}	k_{obs}	U_{obs}	CPU (ms)	L (m^2s)
ContractRF	240	23	0.17966	20	0.16372	9.910	0.01894
ContractRF0.3	240	19	0.15832	17	0.14739	10.960	0.01806
ContractRF3	240	28	0.20554	25	0.19011	9.058	0.02111
Final	240	17	0.14739	28	0.20554	11.694	0.01639
Fixed16	240	17	0.14739	28	0.20554	11.693	0.01639
Fixed2	240	29	0.21063	26	0.19528	8.294	0.02153
Fixed4	240	19	0.15832	19	0.15832	10.845	0.01808
Fixed8	240	18	0.15288	28	0.20554	11.604	0.01667
FlatCART	240	22	0.17438	19	0.15832	9.765	0.01885
FlatCART0.3	240	19	0.15832	17	0.14739	10.186	0.01819
FlatCART3	240	27	0.20042	24	0.18490	8.793	0.02031
FlatRF	240	23	0.17966	20	0.16372	9.929	0.01894
FlatRF0.3	240	18	0.15288	17	0.14739	11.115	0.01727
FlatRF3	240	28	0.20554	25	0.19011	9.075	0.02111
KG0.3	240	19	0.15832	17	0.14739	10.098	0.01828
KG1	240	22	0.17438	19	0.15832	9.745	0.01885
KG3	240	27	0.20042	24	0.18490	8.777	0.02031
OldRAVC	240	26	0.19528	24	0.18490	8.690	0.01961
Random2or4	240	26	0.19528	26	0.19528	9.506	0.02085
Residual	240	29	0.21063	26	0.19528	8.434	0.02152

S4. All-family independent confirmation

Tables include all 20 calibrated policies. Each cell is numerical CPU (ms) / loss (m^2s) / trajectory failures / observation successes. Loads reuse the same inputs and are not additional independent samples.

Table S18: Nominal: $n = 640$ independent inputs.

Policy	Load 1	Load 10	Load 20
ContractRF	10.12/0.00987/24/621	10.22/0.01008/24/621	10.30/0.01040/24/621
ContractRF0.3	11.82/0.00853/20/617	11.95/0.00878/20/617	11.91/0.00947/21/618
ContractRF3	9.28/0.01126/35/611	9.36/0.01143/35/611	9.42/0.01172/35/610
Final	13.31/0.00932/30/581	13.37/0.00984/30/581	13.44/0.01041/30/466
Fixed16	13.31/0.00932/30/581	13.37/0.00984/30/581	13.44/0.01041/30/466
Fixed2	8.37/0.01319/46/600	8.43/0.01350/46/600	8.44/0.01357/46/600
Fixed4	11.23/0.00923/25/603	11.29/0.00970/25/603	11.35/0.01013/25/597
Fixed8	13.17/0.00928/30/583	13.24/0.00977/30/583	13.31/0.01034/30/474
FlatCART	9.70/0.01019/29/617	9.79/0.01038/29/617	9.89/0.01070/29/617
FlatCART0.3	10.74/0.00901/23/611	10.86/0.00923/23/611	10.98/0.00958/23/615
FlatCART3	8.80/0.01107/32/614	8.87/0.01124/32/614	8.94/0.01154/32/614
FlatRF	10.11/0.00987/24/621	10.21/0.01008/24/621	10.30/0.01040/24/620
FlatRF0.3	11.81/0.00853/20/617	11.94/0.00878/20/617	12.07/0.00914/20/617
FlatRF3	9.27/0.01126/35/611	9.34/0.01143/35/611	9.41/0.01172/35/610
KG0.3	10.78/0.00901/23/611	10.86/0.00923/23/611	10.92/0.00956/22/615
KG1	9.72/0.01019/29/617	9.80/0.01038/29/617	9.89/0.01070/29/617
KG3	8.81/0.01107/32/614	8.87/0.01124/32/614	8.95/0.01154/32/614
OldRAVC	8.86/0.01178/35/610	8.92/0.01199/35/610	9.00/0.01230/35/605
Random2or4	9.62/0.01086/32/605	9.68/0.01126/32/605	9.71/0.01147/32/604
Residual	8.45/0.01312/45/602	8.51/0.01325/45/602	8.59/0.01349/45/602

Table S19: Difficult priors: $n = 480$ independent inputs.

Policy	Load 1	Load 10	Load 20
ContractRF	11.76/0.01459/55/435	11.88/0.01513/55/435	11.97/0.01605/55/438
ContractRF0.3	12.72/0.01406/54/434	12.86/0.01461/54/434	12.86/0.01600/56/435
ContractRF3	9.79/0.01510/53/438	9.88/0.01562/53/438	9.95/0.01661/53/438
Final	14.72/0.01870/69/406	14.79/0.02027/69/406	14.87/0.02208/70/266
Fixed16	14.72/0.01870/69/406	14.79/0.02027/69/406	14.87/0.02208/70/266
Fixed2	8.60/0.01740/67/425	8.67/0.01870/67/425	8.67/0.01883/67/424
Fixed4	11.66/0.01455/57/432	11.73/0.01600/57/432	11.80/0.01743/57/426
Fixed8	14.44/0.01787/68/408	14.51/0.01934/68/408	14.59/0.02121/68/269
FlatCART	11.15/0.01461/54/437	11.27/0.01509/54/437	11.39/0.01613/54/437
FlatCART0.3	11.60/0.01432/53/433	11.73/0.01482/53/433	11.86/0.01586/53/436
FlatCART3	9.57/0.01549/59/435	9.66/0.01596/59/435	9.75/0.01697/59/435
FlatRF	11.77/0.01459/55/435	11.89/0.01513/55/435	12.01/0.01612/55/437
FlatRF0.3	12.86/0.01395/54/433	13.00/0.01450/54/433	13.14/0.01552/54/435
FlatRF3	9.80/0.01510/53/438	9.88/0.01563/53/438	9.96/0.01661/53/438
KG0.3	11.35/0.01451/54/436	11.45/0.01499/54/436	11.49/0.01607/54/436
KG1	11.06/0.01463/54/437	11.17/0.01511/54/437	11.27/0.01613/54/437
KG3	9.50/0.01549/59/435	9.58/0.01595/59/435	9.66/0.01696/59/435
OldRAVC	9.58/0.01560/54/438	9.65/0.01658/54/438	9.70/0.01761/54/425
Random2or4	10.05/0.01541/67/427	10.11/0.01678/67/427	10.15/0.01757/67/424
Residual	8.68/0.01690/66/427	8.75/0.01730/66/427	8.82/0.01830/66/427

Table S20: Unfitted geometry: $n = 480$ independent inputs.

Policy	Load 1	Load 10	Load 20
ContractRF	9.89/0.01738/55/434	9.98/0.01748/55/434	10.05/0.01777/55/432
ContractRF0.3	10.91/0.01605/48/436	11.02/0.01618/48/436	11.10/0.01654/48/434
ContractRF3	9.08/0.01817/56/434	9.16/0.01825/56/434	9.23/0.01849/56/433
Final	11.79/0.01346/37/440	11.86/0.01390/37/440	11.92/0.01429/37/407
Fixed16	11.79/0.01346/37/440	11.86/0.01390/37/440	11.92/0.01429/37/407
Fixed2	8.39/0.01877/57/433	8.45/0.01900/57/433	8.46/0.01907/57/433
Fixed4	10.91/0.01565/49/435	10.97/0.01601/49/435	11.03/0.01630/49/430
Fixed8	11.76/0.01355/38/440	11.83/0.01398/38/440	11.89/0.01438/38/408
FlatCART	9.63/0.01641/52/436	9.72/0.01654/52/436	9.82/0.01682/52/435
FlatCART0.3	10.13/0.01575/45/439	10.24/0.01589/45/439	10.35/0.01620/45/436
FlatCART3	8.79/0.01794/58/431	8.87/0.01803/58/431	8.94/0.01828/58/431
FlatRF	9.91/0.01738/55/434	9.99/0.01748/55/434	10.09/0.01774/55/432
FlatRF0.3	11.02/0.01567/46/437	11.13/0.01581/46/437	11.25/0.01613/46/434
FlatRF3	9.10/0.01817/56/434	9.17/0.01825/56/434	9.25/0.01849/56/433
KG0.3	10.01/0.01613/49/437	10.10/0.01626/49/437	10.19/0.01656/49/436
KG1	9.62/0.01641/52/436	9.71/0.01654/52/436	9.80/0.01682/52/435
KG3	8.78/0.01794/58/431	8.85/0.01803/58/431	8.92/0.01828/58/431
OldRAVC	8.80/0.01733/54/436	8.87/0.01744/54/436	8.94/0.01772/54/434
Random2or4	9.58/0.01688/52/436	9.64/0.01717/52/436	9.67/0.01733/52/432
Residual	8.46/0.01877/57/433	8.53/0.01884/57/433	8.60/0.01907/57/433

Table S21: Shifted noise/prior: $n = 360$ independent inputs.

Policy	Load 1	Load 10	Load 20
ContractRF	12.36/0.08499/197/151	12.44/0.08619/197/151	12.46/0.09018/198/154
ContractRF0.3	12.55/0.08499/197/151	12.63/0.08619/197/151	12.64/0.09017/198/154
ContractRF3	11.33/0.08415/195/152	11.39/0.08537/195/152	11.39/0.08938/196/152
Final	17.62/0.10503/217/117	17.67/0.10668/217/115	17.71/0.10826/220/40
Fixed16	17.62/0.10503/217/117	17.67/0.10668/217/115	17.71/0.10826/220/40
Fixed2	8.72/0.10826/204/138	8.75/0.10923/204/138	8.76/0.10930/204/138
Fixed4	11.79/0.08499/197/151	11.83/0.08659/197/151	11.87/0.08815/197/150
Fixed8	16.63/0.09825/208/128	16.67/0.09992/208/128	16.73/0.10182/208/44
FlatCART	11.16/0.08717/196/151	11.23/0.08821/196/151	11.28/0.08921/196/150
FlatCART0.3	11.17/0.08717/196/151	11.23/0.08821/196/151	11.28/0.08920/196/150
FlatCART3	10.47/0.08727/195/147	10.53/0.08830/195/147	10.57/0.08929/195/146
FlatRF	12.39/0.08499/197/151	12.47/0.08619/197/151	12.54/0.08730/197/154
FlatRF0.3	12.63/0.08499/197/151	12.71/0.08619/197/151	12.78/0.08730/197/154
FlatRF3	11.35/0.08415/195/152	11.42/0.08537/195/152	11.47/0.08651/195/152
KG0.3	10.44/0.08562/196/150	10.48/0.08676/196/150	10.50/0.09021/197/149
KG1	10.43/0.08562/196/150	10.48/0.08675/196/150	10.50/0.09021/197/149
KG3	9.81/0.08569/195/146	9.86/0.08683/195/146	9.88/0.09028/196/145
OldRAVC	17.60/0.10508/218/117	17.64/0.10673/218/115	17.69/0.10829/221/43
Random2or4	10.13/0.09422/203/143	10.16/0.09555/203/143	10.18/0.09642/203/143
Residual	12.50/0.07913/187/168	12.55/0.08048/187/168	12.58/0.08174/187/160

S5. Complete routes and additional workload

Route seeds are 5,700,000 + replicate + 10,000 × geometry index (B767: 0; C172: 1; procedural: 2), with replicate 0, 1 or 2. Each raw input is shared by nine variants. Each route has 299 observation opportunities. Means below use three seeds per geometry.

Table S22: B767 means over three independent 600-s routes.

Policy	CPU (s)	RMSE (cm)	L (m^2s)	Ready (%)	Success (%)
ContractRF	2.654	3.962	1.0650	97.77	96.10
Final	3.231	3.860	1.0088	84.17	82.83
Fixed2	2.249	3.968	1.0515	97.55	95.65
Fixed4	3.054	3.971	1.0640	91.30	89.86
FlatRF	2.627	3.963	1.0652	97.77	96.21
KG1	2.664	4.000	1.0713	97.66	96.10
KG1 held	2.876	3.976	1.0656	95.09	93.53
KG1 w/o time	2.645	4.001	1.0717	97.44	95.88
KG1 w/o profile	2.645	4.001	1.0717	97.32	95.76

Table S23: C172 means over three independent 600-s routes.

Policy	CPU (s)	RMSE (cm)	L (m^2s)	Ready (%)	Success (%)
ContractRF	2.527	5.272	2.0817	98.44	89.19
Final	3.035	5.199	2.0319	89.30	82.16
Fixed2	2.232	5.223	2.0324	98.22	89.19
Fixed4	2.711	5.287	2.1030	95.54	86.29
FlatRF	2.511	5.269	2.0793	98.33	89.19
KG1	2.262	5.276	2.0780	98.66	89.19
KG1 held	2.294	5.276	2.0780	98.55	89.07
KG1 w/o time	2.489	5.276	2.0784	98.55	89.19
KG1 w/o profile	2.361	5.276	2.0781	98.77	89.30

Table S24: Procedural means over three independent 600-s routes.

Policy	CPU (s)	RMSE (cm)	L (m^2s)	Ready (%)	Success (%)
ContractRF	2.534	3.932	1.2043	99.22	94.98
Final	2.738	4.127	1.3283	96.43	92.08
Fixed2	2.070	3.921	1.1967	99.22	94.98
Fixed4	2.701	4.080	1.3075	96.32	91.97
FlatRF	2.499	3.932	1.2043	99.00	94.76
KG1	2.491	3.983	1.2383	98.44	94.31
KG1 held	2.434	3.991	1.2430	98.22	94.09
KG1 w/o time	2.391	3.983	1.2379	98.55	94.43
KG1 w/o profile	2.432	3.988	1.2418	98.44	94.20

Table S25: Paired KG1 minus comparator route effects. Descriptive percentile 95% intervals use 10,000 stratified resamples of the nine inputs, three per geometry.

Comparator	Metric	Difference	95% interval
Fixed2	RMSE (cm)	+0.0493	-0.0045, +0.0972
Fixed2	CPU (s)	+0.2885	+0.1693, +0.3918
Fixed2	Success (pp)	-0.0743	-0.8547, +0.7061
Final	RMSE (cm)	+0.0245	-0.0052, +0.0658
Final	CPU (s)	-0.5292	-0.6253, -0.4331
Final	Success (pp)	+7.5065	+5.4998, +9.6618
FlatRF	RMSE (cm)	+0.0320	-0.0154, +0.0822
FlatRF	CPU (s)	-0.0730	-0.1718, +0.0223
FlatRF	Success (pp)	-0.1858	-0.6327, +0.2230
ContractRF	RMSE (cm)	+0.0314	-0.0162, +0.0814
ContractRF	CPU (s)	-0.0993	-0.1897, -0.0261
ContractRF	Success (pp)	-0.2230	-0.7061, +0.2230
KG1 w/o profile	RMSE (cm)	-0.0016	-0.0043, +0.0005
KG1 w/o profile	CPU (s)	-0.0070	-0.0643, +0.0453
KG1 w/o profile	Success (pp)	+0.1115	-0.0372, +0.2601
KG1 w/o time	RMSE (cm)	-0.0001	-0.0009, +0.0006
KG1 w/o time	CPU (s)	-0.0364	-0.1130, +0.0354
KG1 w/o time	Success (pp)	+0.0372	-0.1115, +0.2230
KG1 held	RMSE (cm)	+0.0056	-0.0053, +0.0226
KG1 held	CPU (s)	-0.0625	-0.1802, +0.0385
KG1 held	Success (pp)	+0.9662	+0.2601, +1.5608

Table S26: Additional workload at load 40; reused confirmation inputs, no matching calibration card.

Mixture	Policy	CPU (ms)	L (m^2s)	Traj. fails	Pose success
Nominal	KG1	8.591	0.01393	46	587/640
Nominal	KG1 w/o time	10.041	0.01110	29	580/640
Nominal	Fixed2	8.525	0.01401	46	501/640
Nominal	FlatCART	10.047	0.01111	29	578/640
Nominal	FlatRF	10.369	0.01093	25	577/640
Nominal	ContractRF	8.886	0.01446	47	579/640
Difficult priors	KG1	8.741	0.02105	70	398/480
Difficult priors	KG1 w/o time	11.508	0.01754	54	397/480
Difficult priors	Fixed2	8.759	0.02054	68	329/480
Difficult priors	FlatCART	11.635	0.01754	54	395/480
Difficult priors	FlatRF	12.213	0.01778	55	383/480
Difficult priors	ContractRF	8.898	0.02304	76	385/480
Unfitted geometry	KG1	8.629	0.01964	61	425/480
Unfitted geometry	KG1 w/o time	9.962	0.01711	53	423/480
Unfitted geometry	Fixed2	8.545	0.01942	57	364/480
Unfitted geometry	FlatCART	9.998	0.01712	53	423/480
Unfitted geometry	FlatRF	10.301	0.01806	57	419/480
Unfitted geometry	ContractRF	8.970	0.01975	61	421/480
Shifted noise/prior	KG1	8.537	0.11376	208	121/360
Shifted noise/prior	KG1 w/o time	10.639	0.08992	197	121/360
Shifted noise/prior	Fixed2	8.800	0.11060	204	102/360
Shifted noise/prior	FlatCART	11.414	0.09116	198	121/360
Shifted noise/prior	FlatRF	12.716	0.08981	199	107/360
Shifted noise/prior	ContractRF	8.905	0.11510	210	108/360

Table S27: Additional workload at load 60; reused confirmation inputs, no matching calibration card.

Mixture	Policy	CPU (ms)	L (m^2s)	Traj. fails	Pose success
Nominal	KG1	6.727	0.01908	72	409/640
Nominal	KG1 w/o time	10.236	0.01160	50	384/640
Nominal	Fixed2	8.612	0.01445	48	0/640
Nominal	FlatCART	10.234	0.01159	50	385/640
Nominal	FlatRF	10.559	0.01142	52	301/640
Nominal	ContractRF	6.971	0.01910	73	322/640
Difficult priors	KG1	6.894	0.03460	113	248/480
Difficult priors	KG1 w/o time	11.754	0.01881	78	237/480
Difficult priors	Fixed2	8.852	0.02247	76	0/480
Difficult priors	FlatCART	11.874	0.01881	78	237/480
Difficult priors	FlatRF	12.466	0.01919	85	157/480
Difficult priors	ContractRF	7.153	0.03485	113	165/480
Unfitted geometry	KG1	6.710	0.02200	76	355/480
Unfitted geometry	KG1 w/o time	10.162	0.01747	59	348/480
Unfitted geometry	Fixed2	8.631	0.01976	58	0/480
Unfitted geometry	FlatCART	10.186	0.01747	59	345/480
Unfitted geometry	FlatRF	10.483	0.01839	61	272/480
Unfitted geometry	ContractRF	6.963	0.02204	76	281/480
Shifted noise/prior	KG1	6.869	0.13057	236	59/360
Shifted noise/prior	KG1 w/o time	10.760	0.09231	214	57/360
Shifted noise/prior	Fixed2	8.849	0.11205	208	0/360
Shifted noise/prior	FlatCART	11.532	0.09340	214	57/360
Shifted noise/prior	FlatRF	12.870	0.09205	223	34/360
Shifted noise/prior	ContractRF	7.059	0.13187	238	39/360

S6. Scope audit and reproduction

The release stores raw rolling scans/timestamps, measured deskew motion, priors, future simulation labels, checkpoint features, exports, frozen specifications, 217,080 primary rows, 35,280 workload rows and 81 route traces. Workload rows reuse 1,960 confirmation inputs. A 9,000-request metadata audit uses illustrative seals. These checks are deterministic semantics, not robot failures.

Frozen evidence is checked and regenerated with the supplied scripts.

The analysis programs are `r4_verify.py`, `r4_analyze.py`, `r4_results_author.py` and `r4_figures.py`. Re-running collection, evaluation or routes changes timing and requires rebuilding reports. The bootstrap restores the A19 native dependency from the preserved archive. A SHA256 manifest binds all included files.

A workspace loss removed an earlier R4 working batch. Audited core text was recovered, remaining modules reconstructed, and the entire final batch generated anew with disjoint seeds. No lost-batch outcome enters current tables. Text snapshots and interrupted pilot protocol records are provenance, not pooled evidence. Valid original A19/A20 archives remain byte-exact. The following sections preserve earlier mathematics and conditional deployment references separately.

S7. Secondary event-driven dispatch

The dispatcher forms the compatible policy set from the original 240 simultaneous cards. It minimizes calibration mean loss plus calibration mean CPU, with lexicographic policy-ID tie-breaking. Supported requested allowances are in $[0, 0.15]$. Request events and declared profiles are specified before sensing; the selected policy is constant for the entire declared mixture, not conditioned on individual aircraft identity, scan quality or error. Bounds retain their population scope.

This addition was designed after primary results had been inspected. Its selector uses calibration inputs only and is written to `data/r5_dispatch/selector_freeze.json` before this analysis program reads the existing confirmation records. That ordering does not make the additional question prospective. There are no new geometric samples, executed localization policies or risk cards. Confirmation analysis reuses 640 nominal and 480 procedural inputs across all requests and loads. The selector and every input hash are retained.

Both-marginal requests require each upper bound at most 0.15. The original simultaneous-confidence event supports these two marginal statements together; it does not make the probability of their union at most 0.15. Only $\Pr(E_{\text{traj}} \cup E_{\text{obs}}) \leq \min\{1, U_{\text{traj}} + U_{\text{obs}}\}$ follows without a separately calibrated union event.

The nominal selected pair therefore has union bound 0.19871. No procedural choice has both bounds at most 0.15.

The two comparison interfaces are constructed diagnostics. Policy-level admission reuses a trajectory-certified policy for any requested event, which gives six unsupported assignments among 18 configurations. Checking its actual event cards eliminates those assignments but provides only 12 feasible services. Event-driven selection gives 15. The six unsupported assignments concern evidence applicability; this count does not show that their true risks exceed 0.15.

Table S28: Complete secondary request catalogue. Each marginal allowance is 0.15. Counts enumerate compatible policies, not independent samples or additional statistical tests.

Mixture	α	Request	Eligible	Choice	U_{traj}	U_{obs}
Nominal	1	Trajectory	20	FlatCART3	0.11363	—
Nominal	1	Observation	18	FlatCART3	—	0.08508
Nominal	1	Both marginals	18	FlatCART3	0.11363	0.08508
Nominal	10	Trajectory	20	FlatCART3	0.11363	—
Nominal	10	Observation	18	FlatCART3	—	0.08508
Nominal	10	Both marginals	18	FlatCART3	0.11363	0.08508
Nominal	20	Trajectory	20	FlatCART3	0.11363	—
Nominal	20	Observation	17	FlatCART3	—	0.08508
Nominal	20	Both marginals	17	FlatCART3	0.11363	0.08508
Procedural	1	Trajectory	2	Final	0.14739	—
Procedural	1	Observation	1	FlatRF0.3	—	0.14739
Procedural	1	Both marginals	0	Refused	—	—
Procedural	10	Trajectory	2	Fixed16	0.14739	—
Procedural	10	Observation	1	FlatRF0.3	—	0.14739
Procedural	10	Both marginals	0	Refused	—	—
Procedural	20	Trajectory	2	Fixed16	0.14739	—
Procedural	20	Observation	4	KG0.3	—	0.14739
Procedural	20	Both marginals	0	Refused	—	—

Table S29: Event-selected minus trajectory-selected policy on 480 paired procedural inputs. Percentile 95% intervals are descriptive secondary analyses, without correction for these comparisons.

α	Metric	Difference	95% interval
1	Obs. success (pp)	-0.62500	-2.08333, +0.83333
1	CPU (ms)	-0.77229	-0.93224, -0.61532
1	L ($\mathrm{m}^2\mathrm{s}$)	+0.00221	+0.00056, +0.00421
1	Traj. failures (pp)	+1.87500	+0.41667, +3.33333
10	Obs. success (pp)	-0.62500	-2.08333, +0.83333
10	CPU (ms)	-0.72689	-0.89144, -0.56741
10	L ($\mathrm{m}^2\mathrm{s}$)	+0.00192	+0.00026, +0.00394
10	Traj. failures (pp)	+1.87500	+0.41667, +3.33333
20	Obs. success (pp)	+6.04167	+3.75000, +8.54167
20	CPU (ms)	-1.72638	-1.90481, -1.54998
20	L ($\mathrm{m}^2\mathrm{s}$)	+0.00227	+0.00055, +0.00433
20	Traj. failures (pp)	+2.50000	+1.04167, +4.16667

The dispatch audit executes 97 deterministic cases with illustrative source seals, including changed runtime policy/forecast identities, service/profile/artifact mismatches, invalid numeric fields, expired or wrong-instant observation requests, event substitution, stricter allowance and uncalibrated contexts. No unexpected selection occurs. This is a separate metadata prototype, not an online authorization log from the simulated receiver and not a distribution-shift detector. Its indexed-call benchmark stores all scope and source identifiers in `dispatch_cost.json`: 2,400 process-CPU calls, median 88.76 microseconds and 95th percentile 189.69 microseconds. Runtime/component initialization, scan hashing and transport are excluded. The benchmark is separate from the frozen numerical installation service; integrated receiver timing has not been measured for this dispatcher.

Table S30: Retained complete-route controls, mean of three inputs per aircraft. All use two-second opportunities except Full / 4 s. These are original results, with their original CPU ledger.

Aircraft	Controller	Position RMSE (cm)	CPU (s)
B767	Strict	88.419	4.266
	Tempered	3.437	4.422
	Converged	3.181	8.084
	Full	2.892	16.194
	Full / 4 s	3.225	8.628
	Reuse-DP	3.181	8.610
	Cache removed	3.181	8.360
C172	Strict	2.607	4.726
	Tempered	2.607	4.453
	Converged	2.407	7.737
	Full	2.327	9.003
	Full / 4 s	2.541	5.036
	Reuse-DP	2.600	5.223
	Cache removed	2.600	7.005

Run `python3 source/r5_dispatch.py` to reconstruct selection, audits and paired outcomes, then `python3 source/r5_author.py` to rebuild the tables and risk-feasibility figure. Both programs leave the primary numerical source, forecasts and records unchanged. The primary work–loss figure appears in the main paper.

S8. Retained solver and reuse evidence

The accompanying A19 aircraft release provides five correction actions: Skip, Strict, Tempered, Converged and Full. Strict and Tempered use one update on at most 96 points; Tempered reduces the likelihood weight when a step violates its gate. Converged uses up to 25 iterations on that point limit. Full uses up to 1,500 points and 40 iterations. These controls preserve an important distinction between solver quality, computation reuse and scheduling. The complete original manuscripts, implementations, mesh licenses, protocols and results are included in the archive.

Its source study contains 1,680 distinct seeds and 8,400 paired action outcomes: 480 training, 240 validation, 480 development diagnostics and 480 confirmation sources. The full-route study contains six 1,980-second inputs and 78 localization-service executions, totalling 42.9 simulated route-hours. Table S30 retains the strong controls that determine interpretation. CPU values belong to the original execution environment and are not combined with the new timing ledger.

On C172, fixed-path cache removal repeats 2,941 registrations across 2,967 post-initialization opportunities. It increases estimator CPU from 5.223 to 7.005 s, a 25.4% direct reuse saving. Selected actions and delivered trajectories agree; maximum stored error-trace difference is 1.2×10^{-16}. B767 directly chooses Converged, so cache removal has no equivalent saved registration prefix. The C172 Full / 4 s control is both cheaper and more accurate than Reuse-DP. These retained controls prevent the cache intervention from being interpreted as universal adaptive-policy dominance.

S8.1. The retained finite inner decision layer

For observed probe history h and remaining invocation credit b, let $d_a(h)$ be the unpaid cost of terminal action a. With cached posteriors, $d_a = 0$ for an already available action. The finite controller satisfies

$$R(h,b) = \min_{a:d_a(h)\le b} \{\mathbb{E}[L_a \mid h] + \lambda d_a(h)\}, \tag{S1}$$

$$V(h,b) = \min\left\{R(h,b), \min_{q\notin h:c_q\le b} [\lambda c_q + \mathbb{E}[V(h\oplus(q,Z_q), b-c_q) \mid h]]\right\}. \tag{S2}$$

Three binary acceptance probes have at most $3^3 = 27$ partial histories. With five terminal actions and fixed costs, the query subsets and final action define at most $5 \cdot 2^3 = 40$ candidate budget breakpoints. Between breakpoints, feasible policy trees are unchanged. The optimum is therefore nonincreasing and piecewise constant in budget. At fixed budget it is the minimum of finitely many affine functions of price, hence nondecreasing, concave and piecewise linear.

The two-state noisy-probe example makes the benefit explicit. Equiprobable latent states have losses $(0, 2\mu, \mu - \zeta)$ and $(2\mu, 0, \mu - \zeta)$ for (A, B, F), with costs $(3, 1, 4)$ and budget four. Assume $0 < \zeta < \mu$ and $3\lambda < \zeta$. A cost-three probe executes and caches A and reports the state with symmetric accuracy $r \geq 1/2$. The reuse optimum is

$$V_{\text{reuse}} = \min\{\mu - \zeta + 4\lambda,\ 2\mu(1 - r) + \tfrac{7}{2}\lambda\}. \tag{S3}$$

It strictly improves the prior optimum when $\mu(2r - 1) > \zeta - \lambda/2$. To see this, final F is the best prior action under the price assumption. After probing, cached A and fresh B remain affordable, but F does not. Following the report has expected loss $2\mu(1 - r)$ and work 3.5. Always taking cached A instead costs $\mu + 3\lambda$, exceeding the prior F score. Discarding A removes its affordable branch and cannot improve the prior optimum. This example addresses informational reuse; the new checkpoint interface additionally permits source-version delivery during an ongoing solve.

S9. Source-time latency and conditional error certificates

The scalar correction envelope is retained as an analytic reference. Let source variance be v, process-noise rate q_p, useful horizon T, and action acceptance probability, source contraction and latency be p_a, α_a, τ_a. Successful correction contracts the source uncertainty and then replays process noise. For $0 \leq \tau_a < T$,

$$L_a = Tv + \tfrac{1}{2}q_pT^2 - p_a(1 - \alpha_a)v(T - \tau_a). \tag{S4}$$

Relative to propagation, the priced action is preferable exactly when

$$v > \frac{\lambda c_a}{p_a(1 - \alpha_a)(T - \tau_a)}, \quad p_a > 0, \quad \alpha_a < 1. \tag{S5}$$

Subtracting the two loss expressions proves the boundary. Contracting $v + q_p\tau_a$ would represent information acquired at delivery rather than at the source timestamp. The source-time replay equation in the main manuscript preserves the source-time interpretation.

For a conditional Gaussian task error $e_k \mid \mathcal{H}_k \sim \mathcal{N}(\mu_k, \Sigma_k)$ and tolerated ellipsoid $e_k^\top W e_k \leq 1$, define

$$\beta_k(\delta_k) = \|W^{1/2}\mu_k\| + \sqrt{\lambda_{\max}(W^{1/2}\Sigma_k W^{1/2})\chi^2_{d,1-\delta_k}}. \tag{S6}$$

If $\beta_k \leq 1$ and the history-measurable allowances satisfy $\sum_k \delta_k \leq \Delta$ almost surely, the probability of an event at any specified evaluation instant is at most Δ. Indeed, on the conditional Gaussian ball $\|z\|^2 \leq \chi^2_{d,1-\delta_k}$, the triangle and operator-norm inequalities give $\|W^{1/2}e_k\| \leq \beta_k$. Taking conditional expectations and applying a union bound proves the result without temporal independence. If conditional event probabilities have extra errors ε_k with pathwise sum at most E_{cal}, the bound becomes $\Delta + E_{\text{cal}}$. This certificate requires actual conditional error means and covariances; a registration Hessian alone does not identify them. The executable study uses episode calibration instead.

S10. The exact finite budget model

The preserved numerical allocation study uses 18 stages, 18 credit units, safe uncertainty indices $s = 0, \ldots, 9$ and three geometry states. Its initial state is $(s, g, b) = (6, 1, 18)$. The four abstract actions have costs $(0, 1, 3, 5)$ and successful reductions $(0, 2, 5, 9)$. Corrective acceptance probabilities are

$$P_{g,a} = \begin{bmatrix} 0.18 & 0.38 & 0.62 \\ 0.42 & 0.70 & 0.87 \\ 0.67 & 0.88 & 0.97 \end{bmatrix}. \tag{S7}$$

Before correction, uncertainty grows by one. Acceptance subtracts the action reduction, bounded below by zero; rejection leaves the incremented state. First entry into $s \geq 10$ terminates the model trajectory. Geometry remains unchanged with probability 0.70 and transitions to each other state with probability 0.15. Remaining credit excludes unaffordable actions. These declared kernels define a fully observed finite model.

Let $x_{t,i,a}$ be pre-failure occupancy, $K_{i,a,j}$ its safe transition kernel and $q_{i,a} = 1 - \sum_j K_{i,a,j}$. The standard chance-constrained occupancy program is

$$\min_{x \geq 0} \sum_{t,i,a} c_{i,a} x_{t,i,a}, \tag{S8}$$

$$\sum_a x_{0,i,a} = \nu_i, \qquad \sum_a x_{t+1,j,a} = \sum_{i,a} K_{i,a,j} x_{t,i,a}, \tag{S9}$$

$$\sum_{t,i,a} q_{i,a} x_{t,i,a} \leq \Delta. \tag{S10}$$

The risk sum is exact first-failure probability: its entries correspond to disjoint first-entry events. A feasible policy induces these flows; normalizing feasible flows recovers a randomized Markov policy with identical cost and risk. Credit is part of the state, so every recovered path respects the hard cap. This is an exact model benchmark, not a new general occupancy-equivalence theorem.

At $\Delta = 0.10$, the optimum is 10.3242 expected work units. The optimized mixture of 885 periodic, threshold and geometry rules costs 11.3519. Stronger terminal-aware rules reduce the comparator to 10.9280, and 12-stage rolling dynamic programming gives 10.7898. A standard full-horizon Bellman multiplier mixture reaches the same 10.3242 as the occupancy program. Thus the original 9.05% advantage over stationary-rule mixtures narrows to 5.52% and 4.32% against those stronger controls and disappears against the exact same-information solution. This benchmark illustrates the value of horizon and credit information without assigning its margin to the geometric checkpoint policy.

At 80% nominal correction acceptance, the frozen 0.10 policy's risk rises to 0.2521; reoptimization has minimum risk 0.1601, so 0.10 is infeasible at the original cap in that model. Its high-authority final action is an assumed kernel, not an implemented aircraft recovery operation. The geometric experiments use actual checkpoint candidates and contain no such recovery action.

S11. Conditional deployment reference

A deployment envelope combines a retained simulation anchor s_0 with an independent added airframe-map error of RMS magnitude s_m, delivery delay τ, added position-noise rate q_d and constant motion-bias bound b_d:

$$s_{\text{deploy}}(\tau) = \sqrt{s_0^2 + s_m^2 + q_d\tau + (b_d\tau)^2}. \tag{S11}$$

Variance addition assumes independent, zero-mean added map error; the drift term is a squared envelope contribution. This computes a reference scenario rather than identifying a hardware error model. We use $q_d = 4.8 \times 10^{-5}$ m^2/s and $b_d = \sqrt{3}(0.0015)$ m/s. The anchors are the retained B767 Converged mean 3.181 cm and C172 Full / 4 s mean 2.541 cm, chosen as existing measured-CPU simulation controls.

Table S31: Calculated deployment reference from Eq. (S11). Added map RMS and delay are assumptions. Predicted errors are conditional analytic values, not observed robot errors.

Aircraft	Anchor (cm)	Added map RMS (cm)	Delay (ms)	Predicted RMS (cm)
B767	3.181	1	50	3.34
	3.181	2	150	3.77
	3.181	3	350	4.39
C172	2.541	1	50	2.74
	2.541	2	150	3.25
	2.541	3	350	3.95

S12. Retained experimental figures

The figures in this section restore the original A19 and earlier A20 result displays. Their inputs, solver variants, calibration families and timing ledgers remain those of the retained studies. They provide the detailed evidence underlying the retained-study summaries; their samples and CPU measurements are kept separate from the retained 2,680-episode R4 study. Each original vector PDF is included without changing its plotted data.

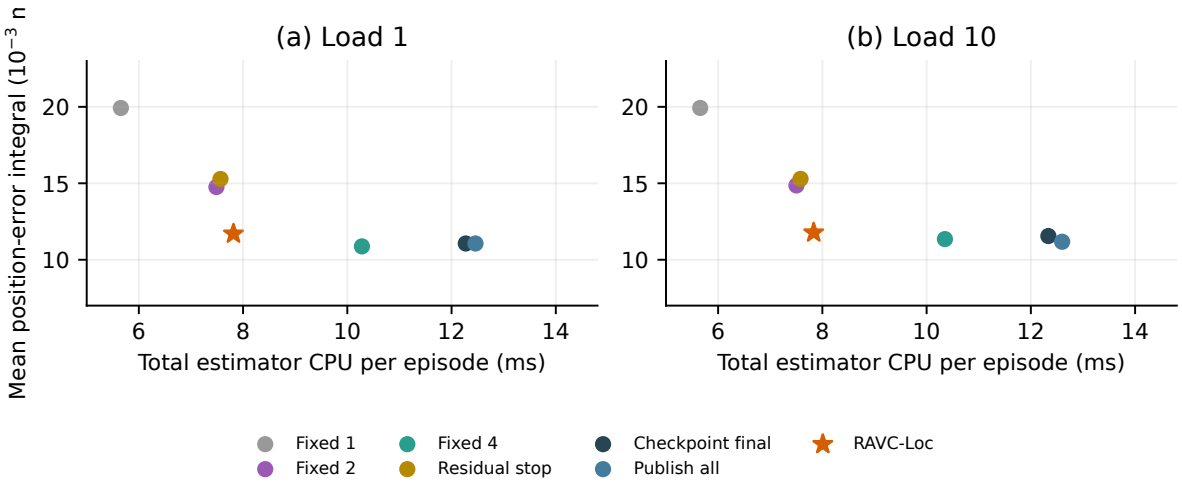


Figure S11: Retained A20 confirmation work–exposure trade-offs at load factors 1 and 10. Each point averages the same 240 independent episodes; the star marks the policy selected using the separate retained calibration set. Positions are descriptive operating points under the original estimator CPU ledger.

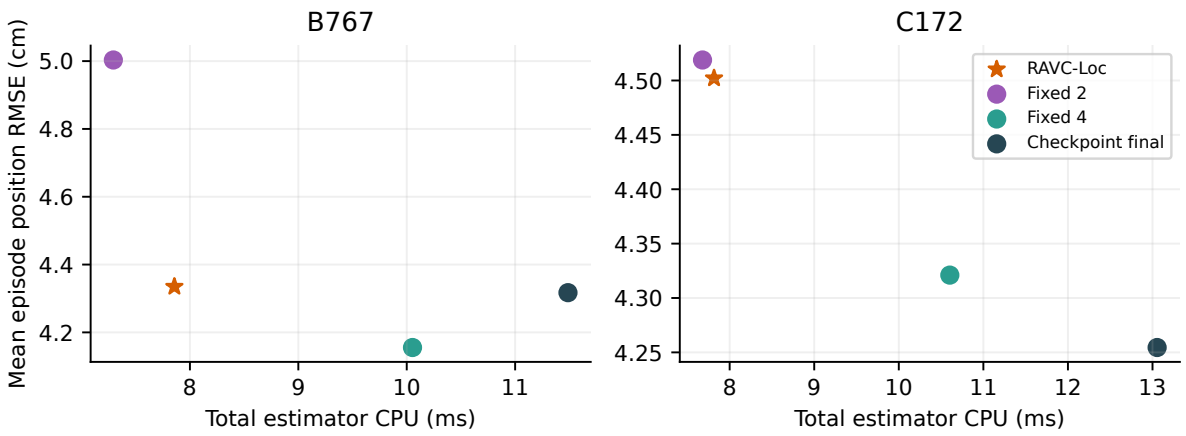


Figure S12: Retained A20 confirmation trade-offs separated by aircraft geometry at load 10. The common retained calibration mixture determines the selected policy. Aircraft subgroups are descriptive and were not used for reselection.

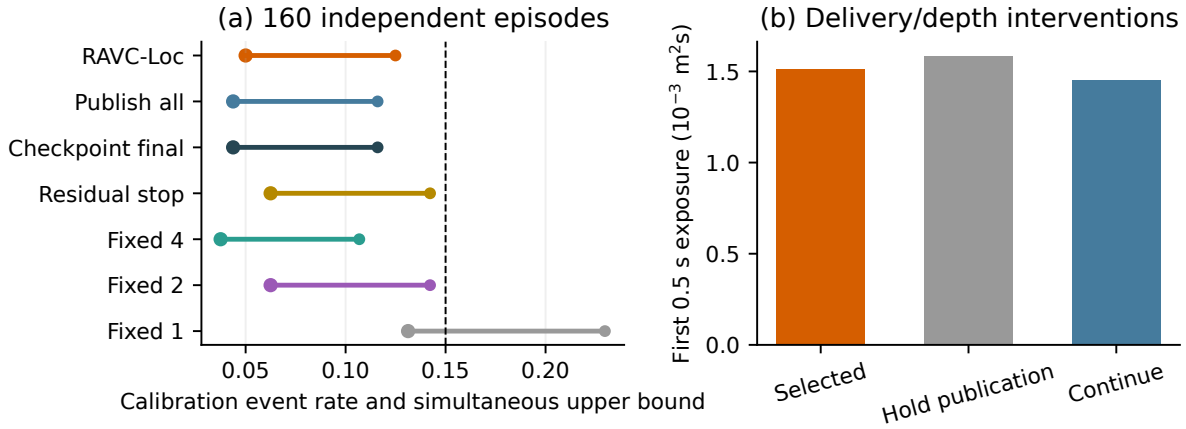


Figure S13: Retained A20 calibration and publication interventions. (a) Point event rates and simultaneous upper bounds for selected controls and RAVC-Loc on 160 independent calibration episodes; the dashed line is $\delta = 0.15$. (b) Startup exposure on 240 confirmation episodes at load 10. Holding publication preserves stopping and the final candidate; continuation preserves publication logic and changes iteration depth.

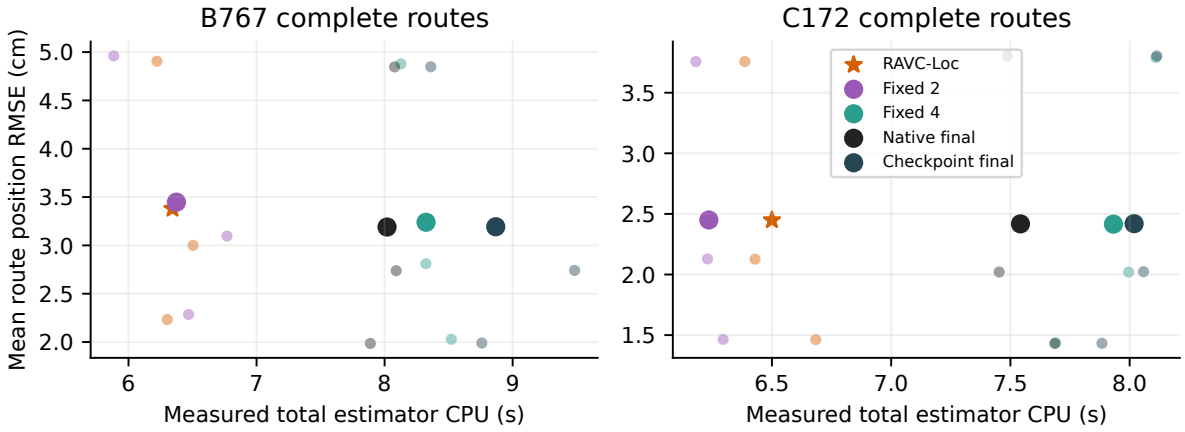


Figure S14: Retained A20 complete-route work–error outcomes. Small points are the three retained realizations per aircraft; large points are their method means over 1,980-s routes. Every control uses the same scan cadence and delivery model within this retained block. The original total estimator CPU ledger is preserved.

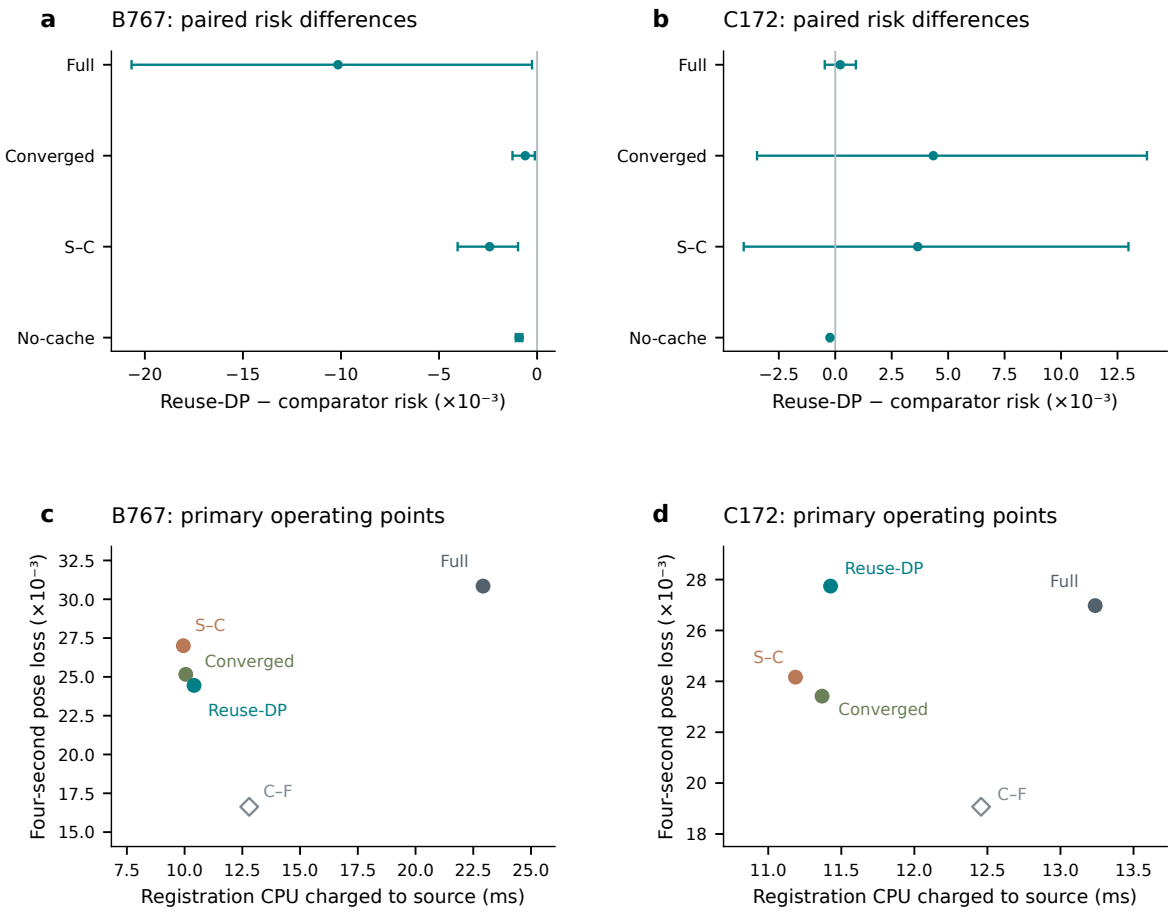


Figure S15: Retained A19 source-level controls on 240 fresh sources per aircraft at $\lambda = 0.3$ and the original primary budget. (a,b) Paired differences in priced risk, Reuse-DP minus the indicated comparator, with pointwise 95% percentile intervals from 4,000 source-bootstrap resamples. (c,d) Mean geometric loss and measured or replay-charged registration CPU. Filled markers indicate feasibility on all observed primary-budget paths; open diamonds mark the relaxed Converged–Full cascade. These action and reuse controls precede the current checkpoint family.

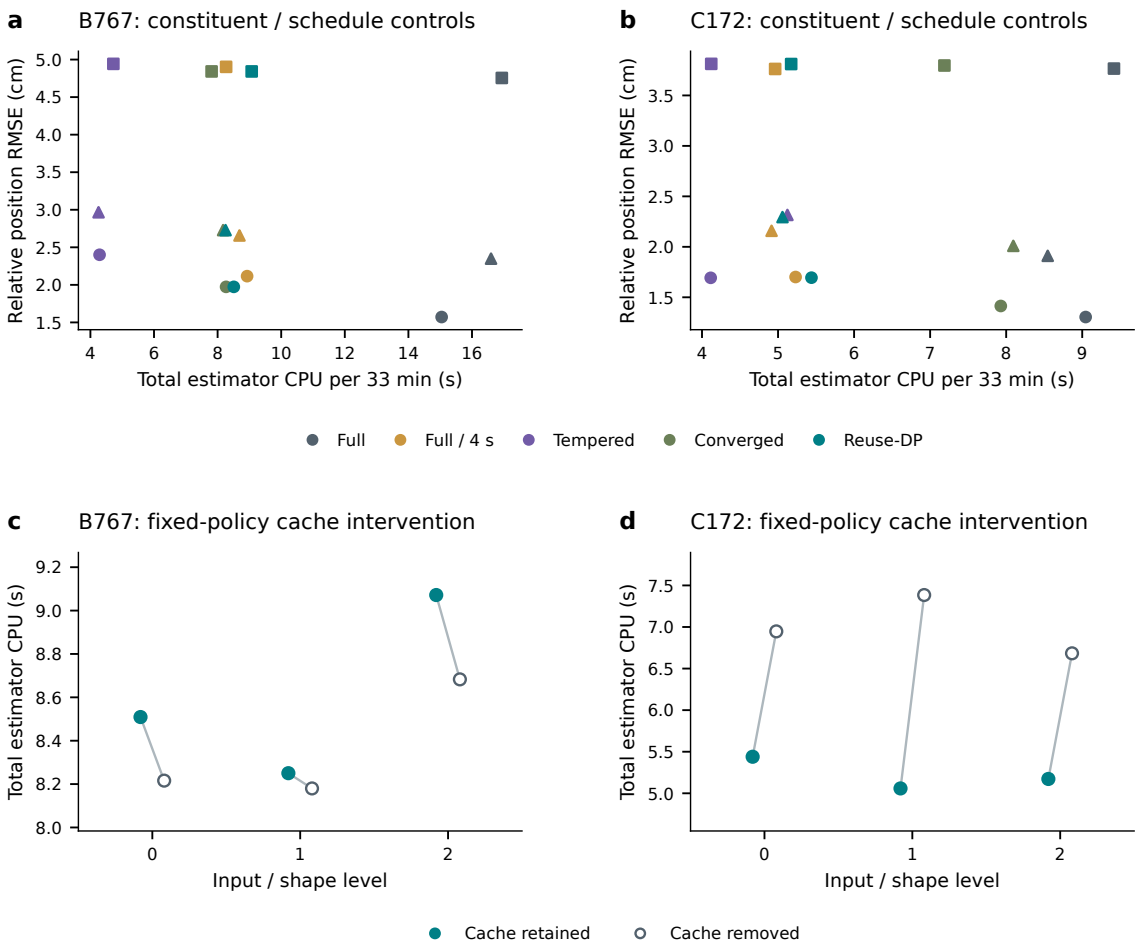


Figure S16: Retained A19 33-minute route controls and cache interventions. Upper panels compare constituent actions and schedules; lower panels remove reuse at fixed policy across the original input/shape levels. The source archive retains all realizations and the separate route CPU accounting.

S13. Additional physical-platform views

Two further original project photographs show platform integration and the aircraft underside. The main R8 paper now includes all four original photographs; these two earlier detail layouts remain for historical continuity. The reproduction bundle preserves the four original JPEGs with their initial dimensions and byte hashes. Photographs provide qualitative engineering context; numerical localization outcomes come from the separately specified simulation inputs.

a Platform integration detail



b Under-fuselage field context



Figure S17: Additional original physical-platform photographs. (a) Side view of platform integration. (b) Field context beneath the aircraft fuselage. Both images retain their original aspect ratios without cropping, pixel retouching, contrast adjustment or generated content.